善用開源軟體及 AI 工具來

妥善進行專案開發

v.20261015

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中原大學資訊工程學系 寅葉電學講堂

2026.10.21

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臺灣主權AI訓練語料庫
(Beta版)

關於語料庫 | 申請須知 | 語料資料集 | 授權條款 | 合作夥伴 | 登入/註冊

臺灣主權AI訓練語料授權條款-第1版

前言：

為促進以人為本的人工智慧（AI）研發與應用，增進公眾利益、提升在地文化價值，並促進自然語料的永續發展，特訂立本授權條款，以加速並提倡相關資料之流通。

一、名詞定義

1. **授權人**：指具備合法權利的個人或單位，依據本授權條款提供其所管理之語料資料予被授權人，以協助被授權人在被明確同意的基礎下，使用該等語料資料於人工智慧的訓練與學習。

https://taic.moda.gov.tw/content/license

趨勢發展務實的角度

探索開源授權與軟體開發間的正確認知

及 AI 工具應用合規之道

大綱

1. 基本提要與開源定義

1.1. 符合開源定義的開源授權條款

1.2. 開源軟體的三大分類與六大特性

2. AI 協作時代的開源應用

2.1. 有了 AI 工具之後，還需要手刻軟體嗎？

2.2. AI 與開源的合作、競爭及其社群衝擊

2.3. 從務實角度得如何因應？

1. 基本提要與開源定義

可以被自由修改、自由散布的軟體專案，就是 / 才是自由開源軟體專案。

A software project is Free and Open Source only if it can be freely modified and redistributed.

Richard Stallman

106 languages

Article Talk Read Edit View history Tools

From Wikipedia, the free encyclopedia

"Stallman" redirects here. For the flutist, see Robert Stallman. For the speculative fiction author, see Robert Lester Stallman.

Richard Matthew Stallman (/ˈ stɔːlmən/ *STAWL-mən*; born March 16, 1953), also known by his initials, **rms**,[1] is an American free software movement activist and programmer. He campaigns for software to be distributed in such a manner that its users have the freedom to use, study, distribute, and modify that software. Software which ensures these freedoms is termed free software. Stallman launched the GNU Project, founded the Free Software Foundation (FSF) in October 1985,[2] developed the GNU Compiler Collection and GNU Emacs, and wrote all versions of the GNU General Public License.

Stallman launched the GNU Project in September 1983 to write a Unix-like computer operating system composed entirely of free software.[3] With that he also launched the free software movement. He has been the GNU project's lead architect and organizer, and developed a number of pieces of widely used GNU software including among

Richard Stallman



https://en.wikipedia.org/wiki/Richard_Stallman



FAIR USE @ https://medium.com/curious-burrows/the-story-of-open-source-so-far-bfcb685d85a4



什麼是 GNU ： https://www.gnu.org.ua/graphics/whatsgnu.zh-tw.html

GNU Project / GNU Linux

GNU is Not Unix

Freeware →← Shareware

Free Software- 四大自由 /The Four Freedoms

Open Source Software- 十項定義 /The Ten Criteria

Open Source Hardware

Open Data

Open Source AI

使用權、修改權、散布權

The right to use, modify, and distribute

源碼／ Source Code

人類能夠理解的程式語言

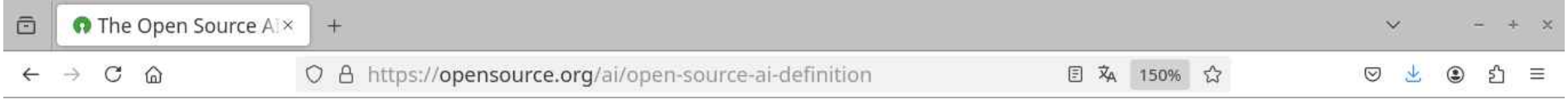
Human-readable programming language

後手可以窺探的程式邏輯

that helps a subsequent party can understand or inspect the Program logic .

Preferred Form to make modifications

原作者自己要修改軟體時，會使用的那一份最適合拿來修改的格式。



open source initiative®

MENU

Preferred form to make modifications to machine-learning systems

The preferred form of making modifications to a machine-learning system must include all the elements below:

- **Data Information:** Sufficiently detailed information about the data used to train the system so that a skilled person can build a substantially equivalent system. Data Information shall be made available under OSI-approved terms.
  - In particular, this must include: (1) the complete description of all data used for training, including (if used) of unshareable data, disclosing the provenance of the data, its scope and characteristics, how the data was obtained and selected,

https://opensource.org/ai/open-source-ai-definition

Source Code → Open Source AI

Preferred form to make modifications

- Data Information: OSI-approved terms
- Code: OSI-approved licenses
- Parameters: OSI-approved terms
- AI model & AI weights included

https://opensource.org/ai/open-source-ai-definition

真正的開源 The real open source =

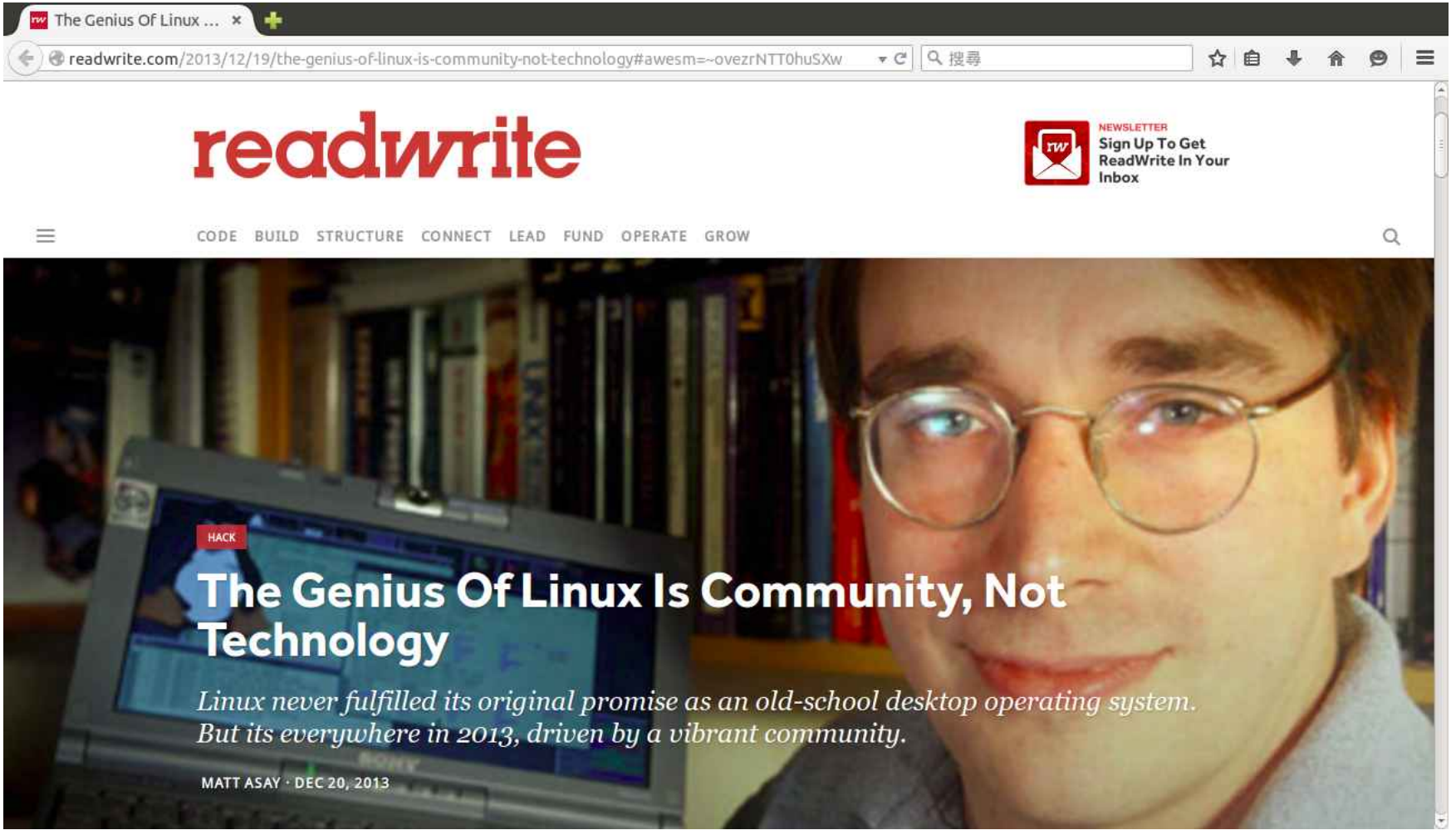
主控權能被後手完整承襲

Full transfer of control to downstream users

基礎建設 as fundamental infrastructure

不是為了商用而啟動開源
而是產生效益後擴大商用

從歷史觀點來說



http://readwrite.com/2013/12/19/the-genius-of-linux-is-community-not-technology#awesm=~ovezrNTT0huSXw

Company	Changes	Total
None	12,550	13.6%
Red Hat	9,483	10.2%
Intel	8,108	8.8%
Texas Instruments	3,814	4.1%
Linaro	3,791	4.1%
SUSE	3,212	3.5%
Unknown	3,032	3.3%
IBM	2,858	3.1%
Samsung	2,415	2.6%
Google	2,255	2.4%
Vision Engraving Systems	2,107	2.3%
Consultants	1,529	1.7%
Wolfson Microelectronics	1,516	1.6%
Oracle	1,248	1.3%
Broadcom	1,205	1.3%

Company	Changes	Total
NVidia	1,192	1.3%
Freescale	1,127	1.2%
Ingics Technology	1,075	1.2%
Renesas Electronics	1,010	1.1%
Qualcomm	965	1.0%
Cisco	871	0.9%
The Linux Foundation	840	0.9%
Academics	831	0.9%
AMD	820	0.9%
Inktank Storage	709	0.8%
NetApp	707	0.8%
LINBIT	705	0.8%
Fujitsu	694	0.7%
Parallels	684	0.7%
ARM	664	0.7%

http://readwrite.com/2013/12/19/the-genius-of-linux-is-community-not-technology#awesm=~ovezrNTT0huSXw

Open Source

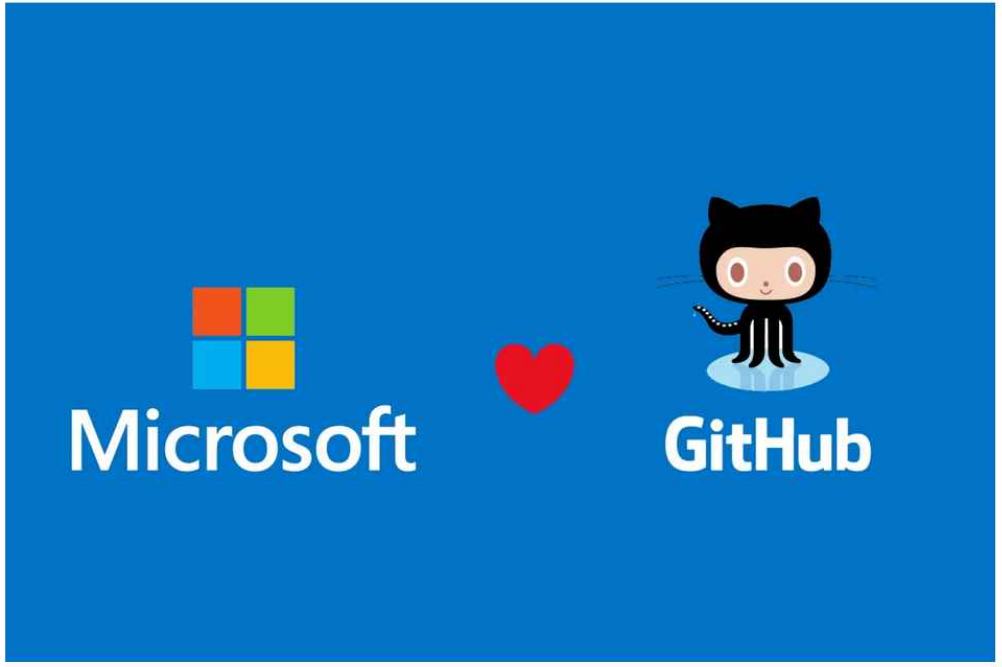
當代跨界合作的有效手段

Ecosystem

建構開放標準統合
軟硬整合的生態系

TECH MICROSOFT

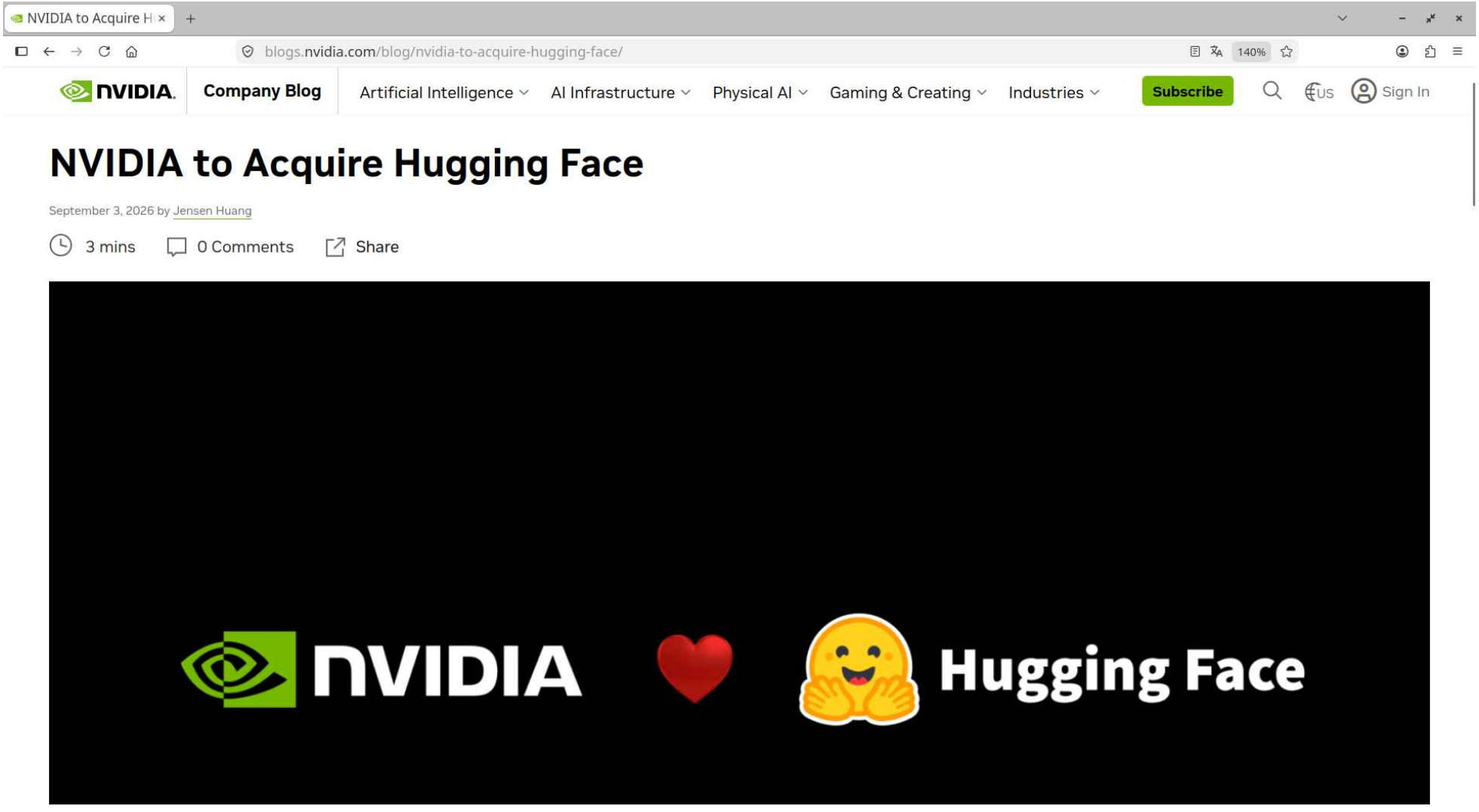
Microsoft confirms it's acquiring GitHub for $7.5 billion



by Tom Warren
Jun 4, 2018, 9:02 PM GMT+8

0 Comments

https://www.theverge.com/2018/6/4/17422788/microsoft-github-acquisition-official-deal



https://blogs.nvidia.com/blog/nvidia-to-acquire-hugging-face/



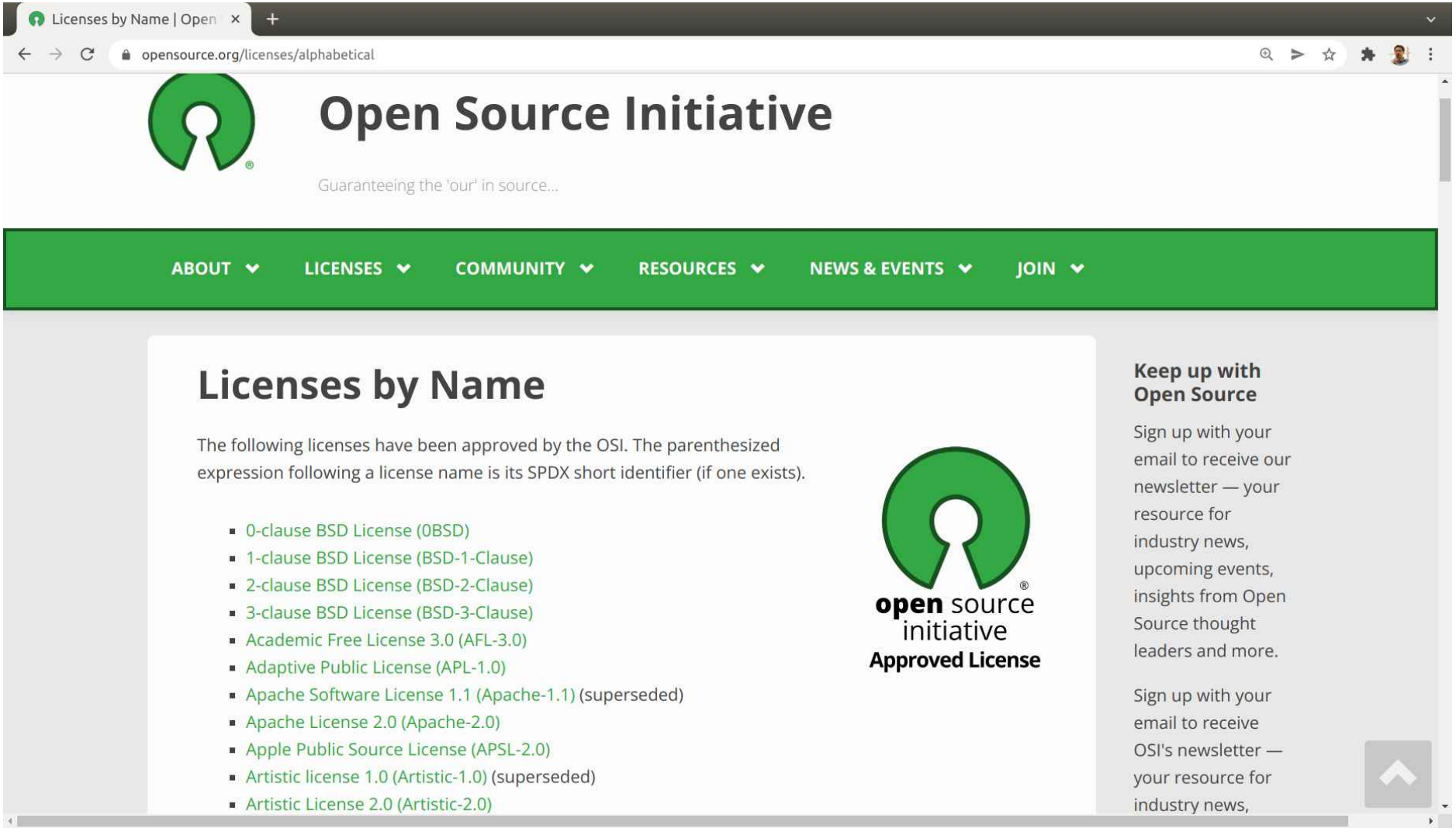
https://www.youtube.com/watch?v=xPh-5P4XH6o



https://www.youtube.com/watch?v=xPh-5P4XH6o

可以被自由修改、自由散布，沒有使用期間、範圍、目的限制的軟體專案，就是 / 才是真正的自由開源軟體專案。

1.1. 符合開源定義的開源授權條款



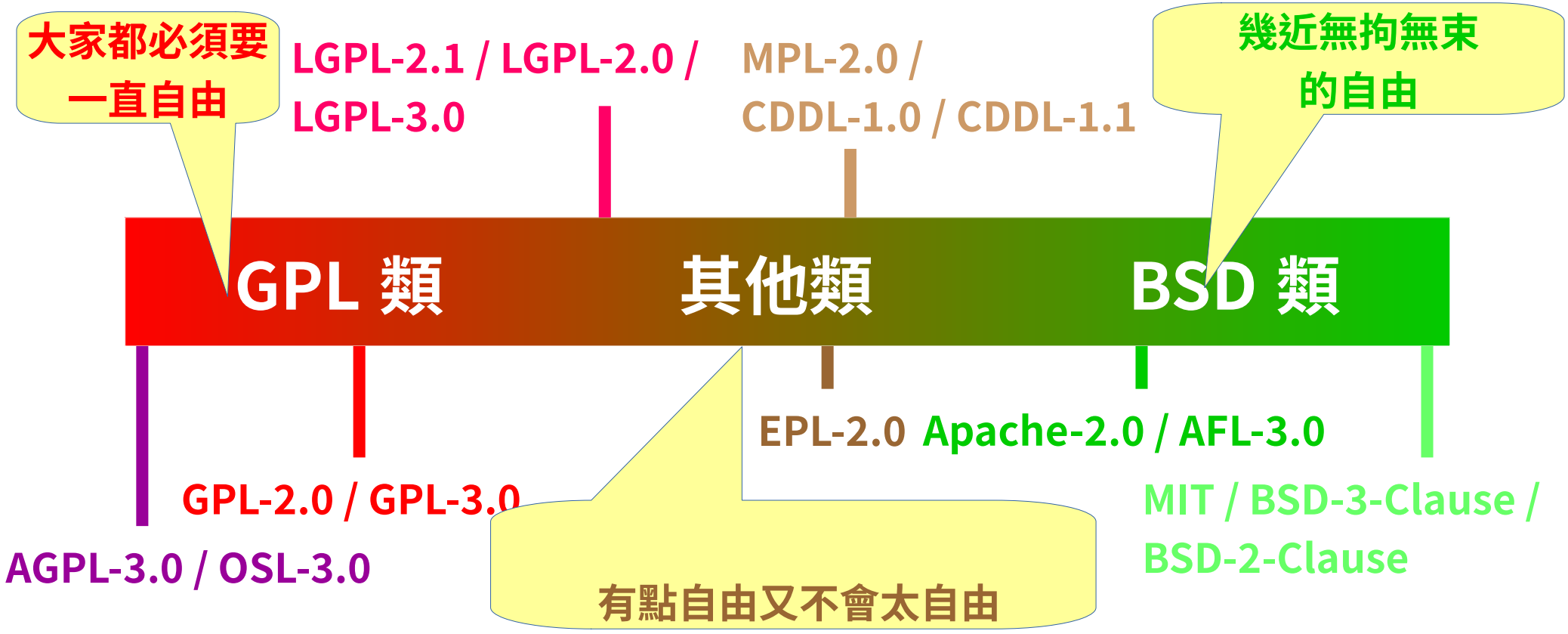
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OSI-approved Licenses sorted by name by the date 2021.10.18: https://opensource.org/licenses/alphabetical

1.2. 開源軟體的三大分類與六大特性

開源軟體的三大分類


大家都必須要
一直自由
LGPL-2.1 / LGPL-2.0 /
LGPL-3.0
MPL-2.0 /
CDDL-1.0 / CDDL-1.1
幾近無拘無束
的自由
GPL 類
其他類
BSD 類
AGPL-3.0 / OSL-3.0
GPL-2.0 / GPL-3.0
EPL-2.0
Apache-2.0 / AFL-3.0
MIT / BSD-3-Clause /
BSD-2-Clause
有點自由又不會太自由

開源軟體的六大特性

開源授權的六大特性

符合 FSF 自由軟體定義或 OSI 開放源碼定義者

1 、非專屬授權 non-exclusive

2 、不可撤回 irrevocable

3 、免授權金 royalty-free

4 、不附隨擔保 no warranties

5 、不限制目的、時間、地域

for any purpose, perpetual, worldwide

6 、使用、重製、修改、發布

freedom to run, copy, modify, and redistribute

1. 非專屬授權 non-exclusive

2. 不可撤回 irrevocable

3. 免授權金 royalty-free

4. 不附隨擔保 no warranties

3．免授權金 royalty-free

5．不限制目的、時間、地域

for any purpose, perpetual, worldwide

6. 使用、重製、修改、發布

freedom to run, copy, modify, and redistribute

開源軟體：

1. 只要合規、取得的開源授權永久不會撤回
2. 使用規則就是依取得時那份條款
3. 要求註引出處、或必要時相同方式分享，但原則上不會有其他限制。

沒有比（真正的）開源更浪漫的事……

Love is not love　愛非愛
Which alters when it alteration finds,　若它一變即改
Or bends with the remover to remove.　或因對方離開，也就離開
O no! It is an ever-fixèd mark　愛是永恆不移的標記
That looks on tempests and is never shaken.　縱處動盪也不動搖

—William Shakespeare, Sonnet 116: “Let Me Not to the Marriage of True Minds”



開源授權條款為骨、開源程式碼協作為肉，整體才能建構可維運的共生生態系。

2. AI 協作時代的開源應用

2.1. 有了 AI 工具之後，還需要手刻軟體嗎？

Crafting Apps: open

getartcraft.com/apps

130%

HOME IMAGE VIDEO RESOURCES DOWNLOAD SUPPORT CRAFT APPS PRICING LAUNCH APP

01 / CRAFTING APPS

THIRTEEN APPS · OPEN SOURCE · PURE RUST

CRAFTING APPS

Thirteen apps. One craft.

Image editing, vector illustration, video, photography, PDFs, motion graphics, page layout, interface design, word processing, audio, presentations, CAD and spreadsheets. Native, open-source apps from the ArtCraft team, built in Rust and free to use.



DOWNLOAD LAUNCHER

JOIN THE DISCORD

BROWSE ON GITHUB

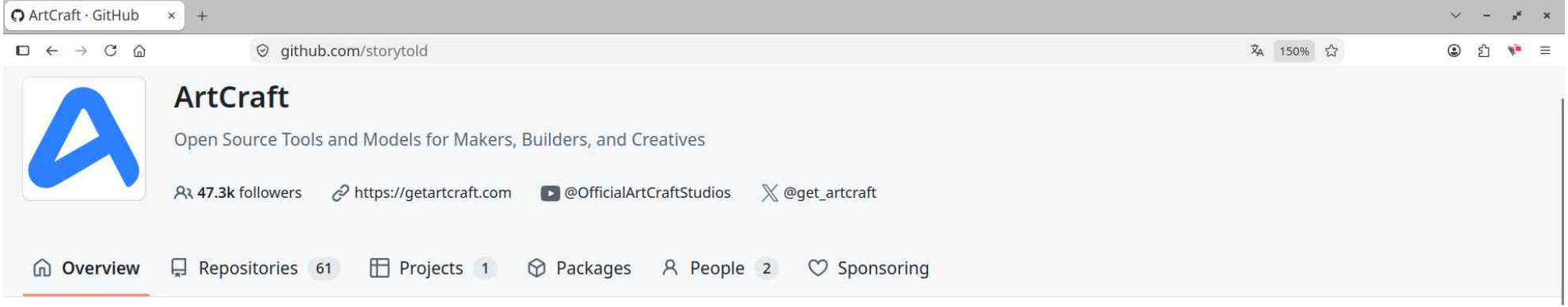
FOR LINUX · FLATPAK (X86_64)

Recommended: ArtCraft Launcher installs every app below and keeps them all up to date. Each app also has its own installers on its page. Learn more

SHARE COPY LINK X REDDIT BLUESKY LINKEDIN

LINEUP
PRINCIPLES
GET STARTED

https://getartcraft.com/apps



https://github.com/storytold

IF IT QUACKS LIKE ADOBE

"Software is over": Bold AI developer takes aim at Adobe with open source clones

Opus-built Creative Cloud alternatives are ambitious, free, and nowhere near finished.

KYLE ORLAND – 2026年10月8日 上午5:56 | 399



Think that's a screen from Photoshop? Look again... Credit: Artcraft

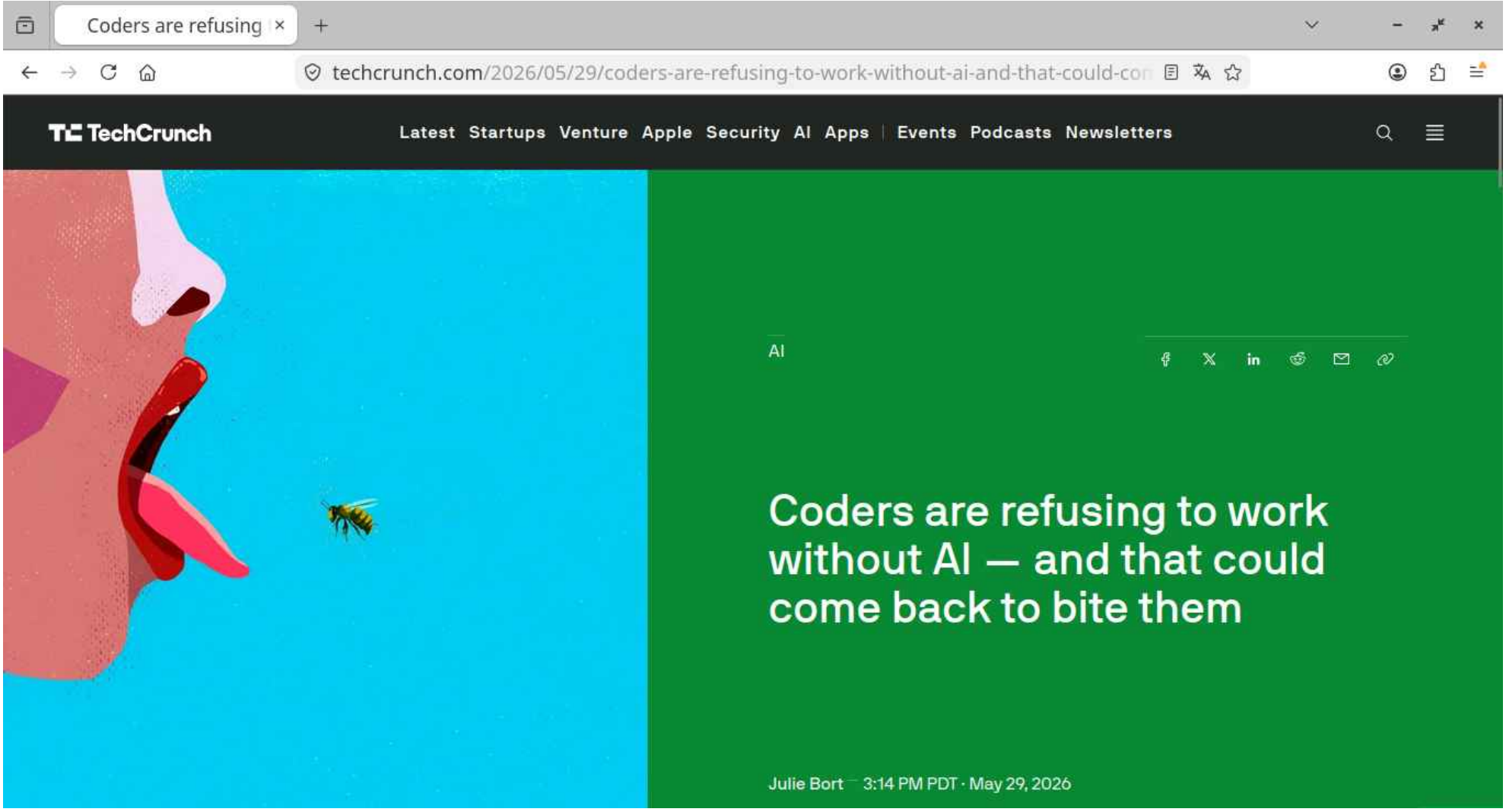
For a while, those hoping to avoid the difficulty and expense of dealing with Adobe's Creative Suite have been able to use several free and/or open source alternatives that replicate some or all of those capabilities. Now, one developer is using AI-powered reverse engineering to mimic the look, feel, and functionality of Adobe's well-known software with what he hopes will soon be perfect fidelity.

https://arstechnica.com/ai/2026/10/software-is-over-bold-ai-developer-takes-aim-at-adobe-with-open-source-clones/

這些事是怎麼做的？

Vibe Coding

（氛圍編程 / 感覺編程）



https://techcrunch.com/2026/05/29/coders-are-refusing-to-work-without-ai-and-that-could-come-back-to-bite-them/

METR （Model Evaluation & Threat Research）

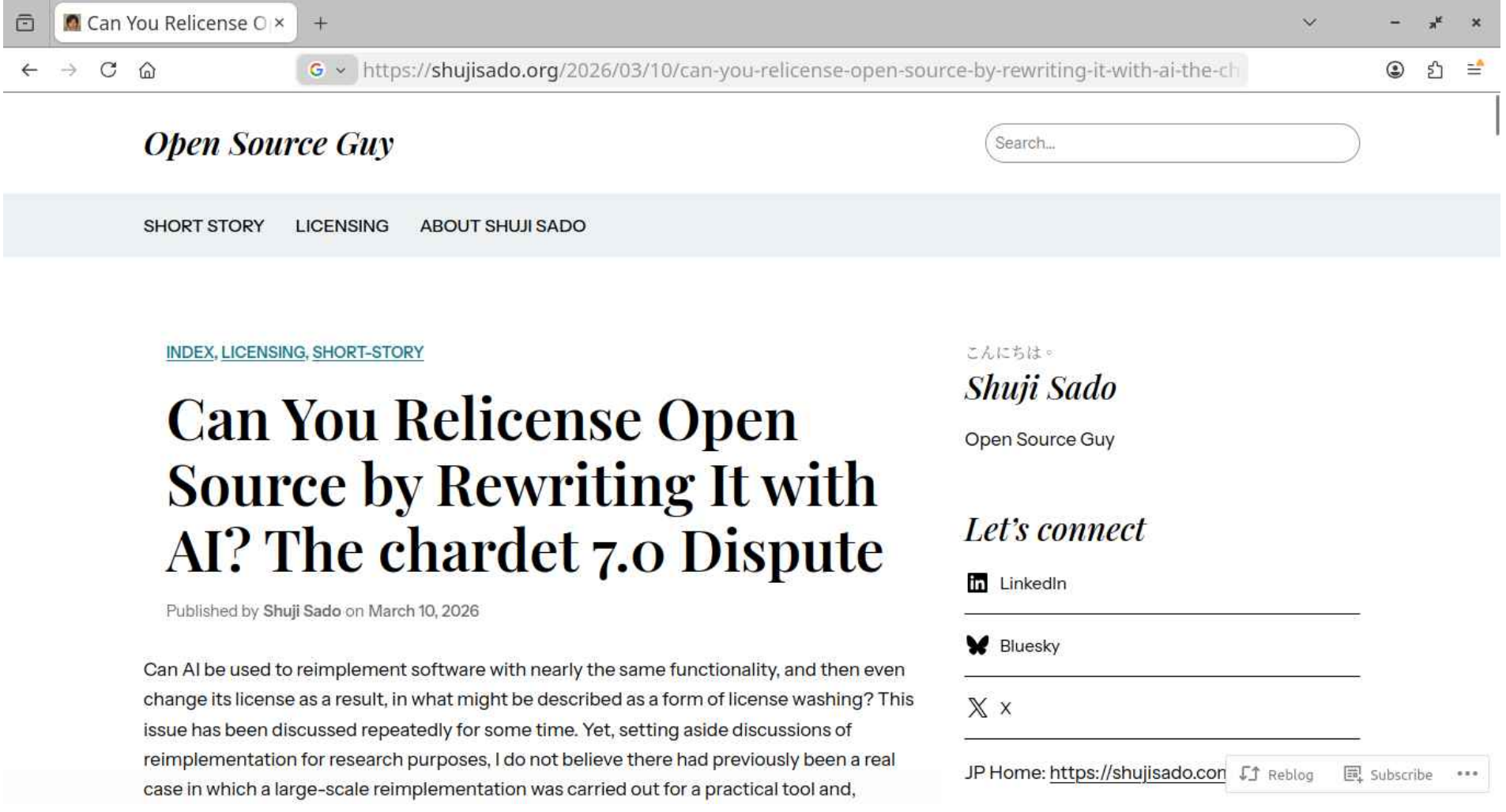
- AI 加快了程式開發？
- 需注意之處：

1.程式碼品質低落問題

2.Bug 或資安漏洞一致性

3.長期 technical debt 的積累 → 技術債

4.開發者對底層原理理解下降



https://shujisado.org/2026/03/10/can-you-relicense-open-source-by-rewriting-it-with-ai-the-chardet-7-0-dispute/

Python chardet 7.0 事件

1.長年採用 LGPL

2.維護者 Dan Blanchard 宣布以 Claude Code 協助進行「ground-up rewrite」

3.隨後將 7.0 版改採 MIT License

4.保留同一個專案名稱、同一個 API 、同一個套件位置

5.引發是否屬於合法轉換授權與尊重開源方法論的爭議



What does the MIT license cover in chardet?

New issue

#334

Closed

#349



richardfontana opened on Mar 7

In #327 **@dan-blanchard** says:

> I did not write the code by hand, but I was deeply involved in designing, reviewing, and iterating on every aspect of it. [. . .] But the evidence here is clear: 7.0 is an independent work, not a derivative of the LGPL-licensed codebase. The MIT license applies to it legitimately.

The current LICENSE file containing the text of the MIT license is prefixed by "Copyright (c) 2024 Dan Blanchard"

Assignees

No one assigned

Labels

No labels

Type

No type

Fields Give feedback

https://github.com/chardet/chardet/issues/334

為什麼可以這樣做？

Clean Room Engineering

(淨室工程)

WineHQ - Run Wind × Clean Room Guideli × Clean-room design - ×
en.wikipedia.org/wiki/Clean-room_design 133%
WIKIPEDIA 25 years of the free encyclopedia
Donate Create account Log in

Clean-room design

10 languages

Article Talk Read Edit View history Tools

From Wikipedia, the free encyclopedia

(Redirected from Clean room design)

> *This article is about the design-copying technique. For the manufacturing or research environment, see clean room. For the software development methodology, see cleanroom software engineering.*

Clean-room design (also known as the Chinese wall technique) is the method of copying a design by reverse engineering and then recreating it without infringing any of the copyrights associated with the original design. Clean-room design is useful as a defense against copyright infringement because it relies on independent creation. However, because independent invention is not a defense against patents, clean-room designs typically cannot be used to circumvent patent restrictions.

The term implies that the design team works in an environment that is "clean" or demonstrably uncontaminated by any knowledge of the proprietary techniques used by the competitor.

https://en.wikipedia.org/wiki/Clean-room_design

軟體程式碼

為著作權保護之客體

著作權保護標的

僅止於著作的表達（形式）

著作權法

並不保護「**點子**」！

除非您去申請專利或是把它當營業秘密！

著作權法第 10-1 條：

http://law.moj.gov.tw/LawClass/LawSingle.aspx?Pcode=J0070017&FLNO=10-1

著作權法第 10-1 條

依本法取得之著作權，其保護僅及於該著作之表達，而不及於其所表達之思想、程序、製程、系統、操作方法、概念、原理、發現。

https://law.moj.gov.tw/LawClass/LawSingle.aspx?pcode=J0070017&flno=10-1

17 U.S. Code § 102(b) – Subject matter of copyright: In general

In no case does copyright protection for an original work of authorship extend to any idea, procedure, process, system, method of operation, concept, principle, or discovery, regardless of the form in which it is described, explained, illustrated, or embodied in such work.

https://www.govinfo.gov/content/pkg/USCODE-2021-title17/html/USCODE-2021-title17-chap1-sec102.htm



頭條要聞 〉 北捷封殺12款App 網友怒

分享到FB 分享到g+ 分享到Plurk 分享到Twitter

北捷封殺12款App 網友怒

「正版太爛 好用的卻被下架」

2013年01月16日 g+1 29



台北捷運公司要求侵權的App下架。廖祥瑞攝

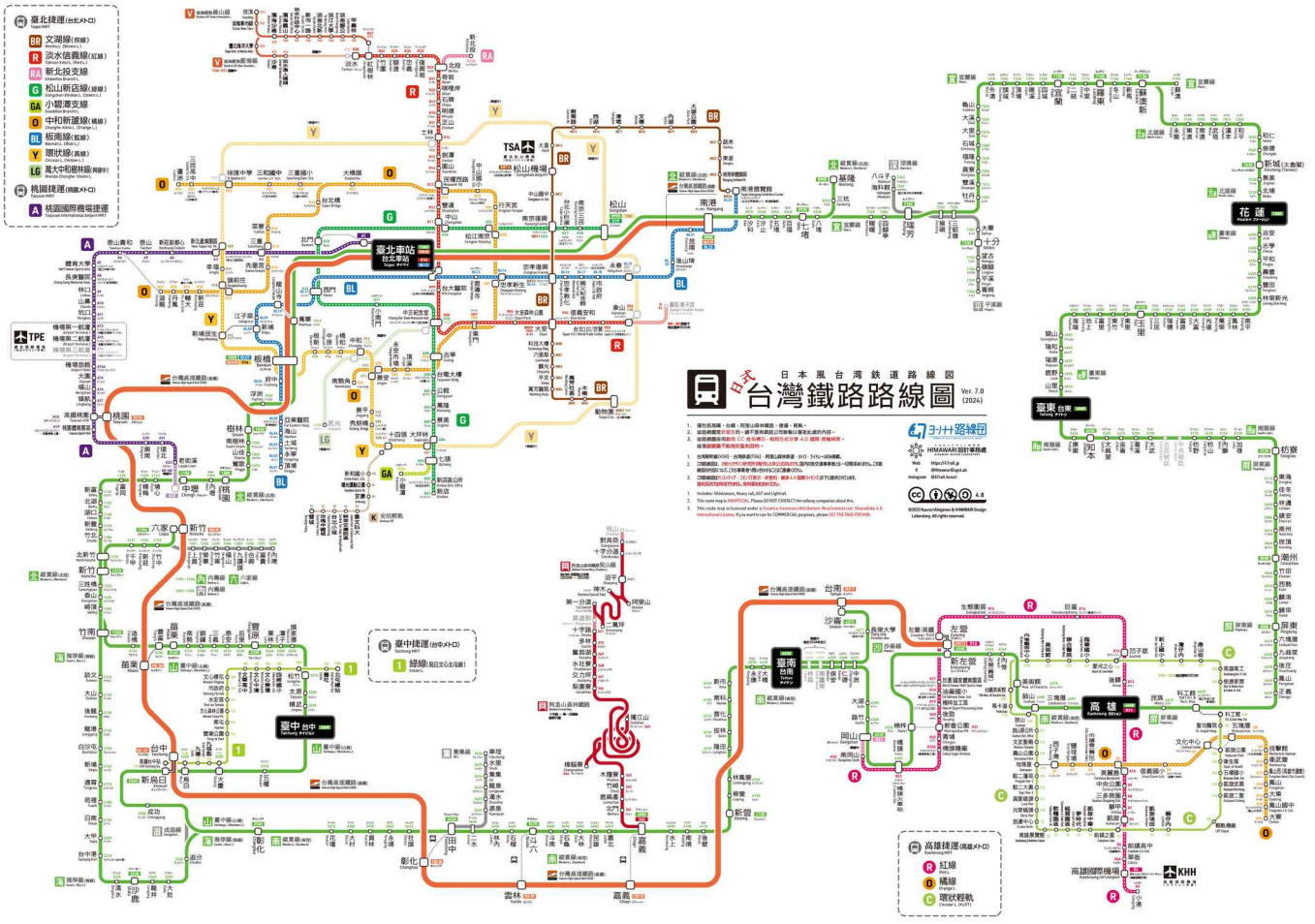
【許麗珍／台北報導】台北捷運公司向手機軟體Ａｐｐ業者開鍘，去年底以違反著作權為由，發文要求Google Play與App Store兩下載平台撤除十二款未經北捷授權的捷運網圖Ａｐｐ，截至昨日已有五款下架，並籲民眾下載北市交通局正版「台北好行」Ａｐｐ。網友砲轟「台北好行」太爛，好用Ａｐｐ卻被要求下架。

按讚看蘋果 食安問題追追追

網友kenf（肯夫）在台大PTT（telnet://ptt.cc）貼文，指Google Play知名北捷查詢Ａｐｐ「台北捷運地圖」遭下架，檢舉的北捷並對涉及侵權的其他同類型十一款Ａｐｐ，向軟體商店提出檢舉，引發網友熱烈討論。

Fair Use at: http://web.metro.taipei/img/all/metrotaipeimap.jpg

Public Domain for the text at law at: http://www.appledaily.com.tw/appledaily/article/headline/20130116/34773133/



HIMAWARI 設計事務處， " 日式台灣鐵路路線圖 "--2024 Ver.7.0 @ https://47rail.jp/get-routemap/taiwan/, CC-BY-NC-SA 4.0 .





左圖： Metro Taipei 官方路網圖，右圖：依左圖採用 OpenAI ChatGPT-6.1 Sol 轉換為 Tokyo Metro 風格，皆採合理使用 (FAIR USE) 立場引用。



Taipei_Metro_Stations_and_Connections.md — ReText

檔案 編輯 說明

預覽 樣式 特殊符號

Taipei_Metro_Stations_and_Connections ×

臺北捷運路網圖：站名與連線列表

> 依使用者指定圖片逐線整理。本文是圖片的結構化轉錄，非截至今日的營運路網查核。圖中淡色路線也納入，但顏色深淺不單獨當作營運狀態證據。

讀法與範圍

- 每個路線表依站序排列；相鄰兩列代表同一路線直接相連，原則上可雙向理解。
- 表中的「相鄰站」只計同一路線的邊，不含步行轉乘。
- 分支另列，不把兩支線的終點誤接成同一路段；分支節點為北投、七張、大橋頭、濱海沙崙。
- 保留圖上站名用字（如「台北車站」「台北101/世貿」）；不包含站名旁的學校、醫院等副站名。
- 站碼將圖上分行字母與數字合併，例如 BL／01 寫成 BL01。
- 台鐵、高鐵僅作轉乘標記；原圖未提供其完整站序，故不推補鐵路連線。

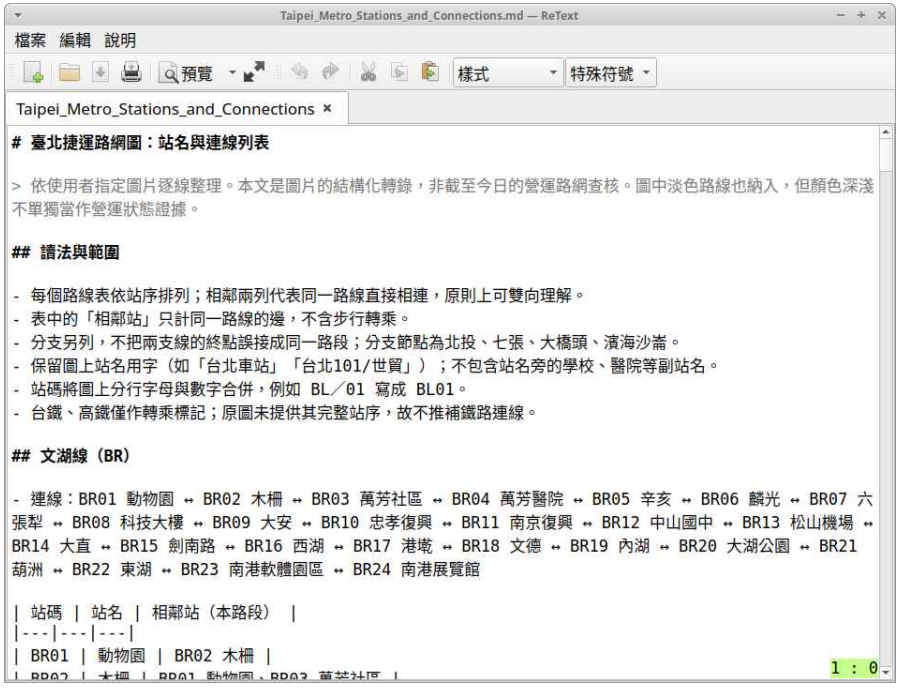
文湖線（BR）

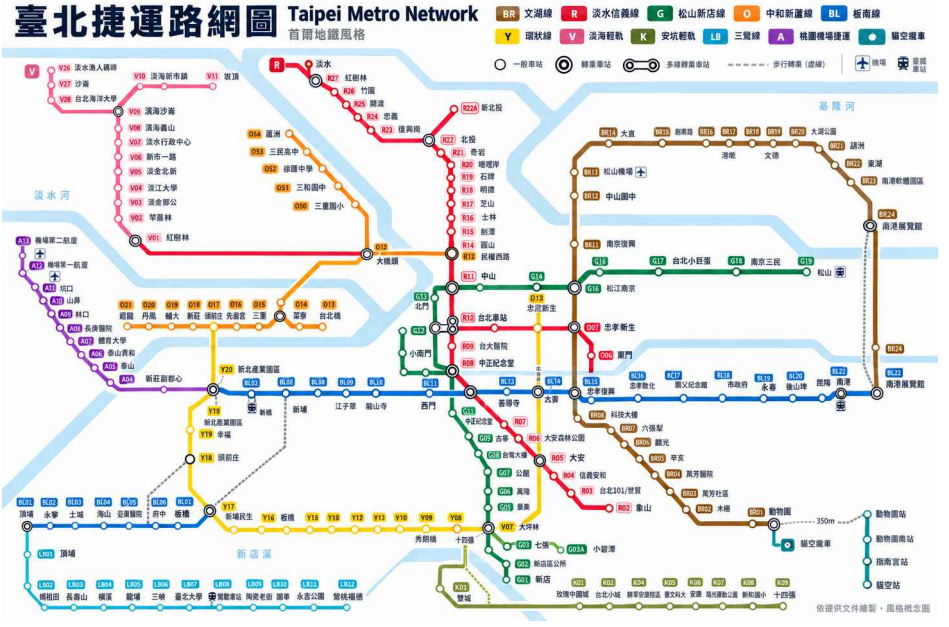
- 連線：BR01 動物園 ↔ BR02 木柵 ↔ BR03 萬芳社區 ↔ BR04 萬芳醫院 ↔ BR05 辛亥 ↔ BR06 麟光 ↔ BR07 六張犁 ↔ BR08 科技大樓 ↔ BR09 大安 ↔ BR10 忠孝復興 ↔ BR11 南京復興 ↔ BR12 中山國中 ↔ BR13 松山機場 ↔ BR14 大直 ↔ BR15 劍南路 ↔ BR16 西湖 ↔ BR17 港墘 ↔ BR18 文德 ↔ BR19 內湖 ↔ BR20 大湖公園 ↔ BR21 葫洲 ↔ BR22 東湖 ↔ BR23 南港軟體園區 ↔ BR24 南港展覽館

站碼	站名	相鄰站（本路段）
BR01	動物園	BR02 木柵
BR02	木柵	BR01 動物園、BR03 萬芳社區

1 : 0

左圖： Metro Taipei 官方路網圖、 FAIR USE ，右圖：將左圖透過 OpenAI ChatGPT-6.1 Sol 摘取其各站相連之事實資訊。





左圖： OpenAI ChatGPT-6.1 Sol 摘取 Metro Taipei 官方路網圖各站相連之事實資訊文件；
右圖：將左圖文件提供予 OpenAI ChatGPT-6.1 Sol 依首爾地鐵風格繪製之路網圖。

如果霍格華茲有台灣生：

1. 鹹酥雞珍奶 2. 塑膠掃帚 3. 大同電鍋

4. 藍白拖 5. 擲筊 puá̍h-pue

https://twitter.com/I_dot_car/status/1572404697860018179





FAIR USE

左圖：桃色壞男孩 Peachy Boys @ https://tinyurl.com/39j4p2zc

右圖：馬鈴熊 PotaBear @ https://www.instagram.com/pota_bear_ti/

著作權法保護的標的為「表達的形式」

不及於背後的概念、思想

Coding Style, Painting Style, Writing Style

不受著作權保護、技法學習是能被接受的 ○

但累積式的抄寫程式碼或文句、圖畫片段 ×



https://twitter.com/I_dot_car/status/1572404697860018179

AI 重作者主張學習事實資訊

沒有著作權上的抄襲

抄 – 重製權、襲 – 改作權：針對既有表達

(1) 接觸

(2) 實質相似

最高法院 81 年台上字第 3063 號判決。

認定抄襲之要件有二，即（一）接觸，（二）實質相似。

成功的 CLEAN ROOM

(1) 不要接觸表達、只接觸概念或邏輯

(2) 避免最終產出實質近似

Substantial similarity

Crafting Apps: open

getartcraft.com/apps

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01 / CRAFTING APPS

THIRTEEN APPS · OPEN SOURCE · PURE RUST

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DOWNLOAD LAUNCHER

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FOR LINUX · FLATPAK (X86_64)

Recommended: ArtCraft Launcher installs every app below and keeps the all up to date. Each app also has its own installers on its page. Learn more

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LINEUP
PRINCIPLES
GET STARTED

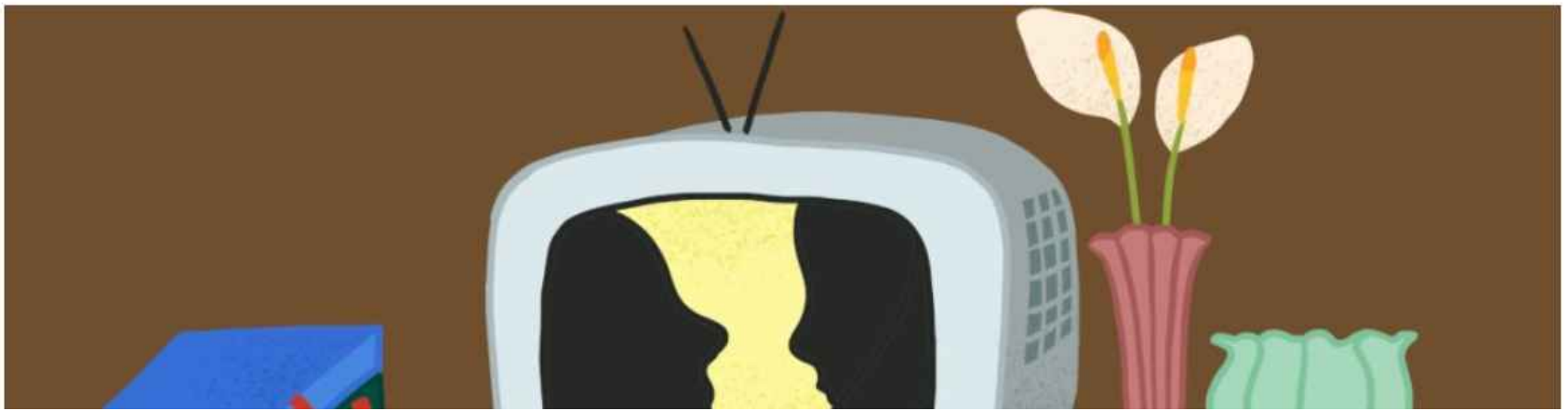
https://getartcraft.com/apps

2.2. AI 與開源的合作、競爭及其社群衝擊

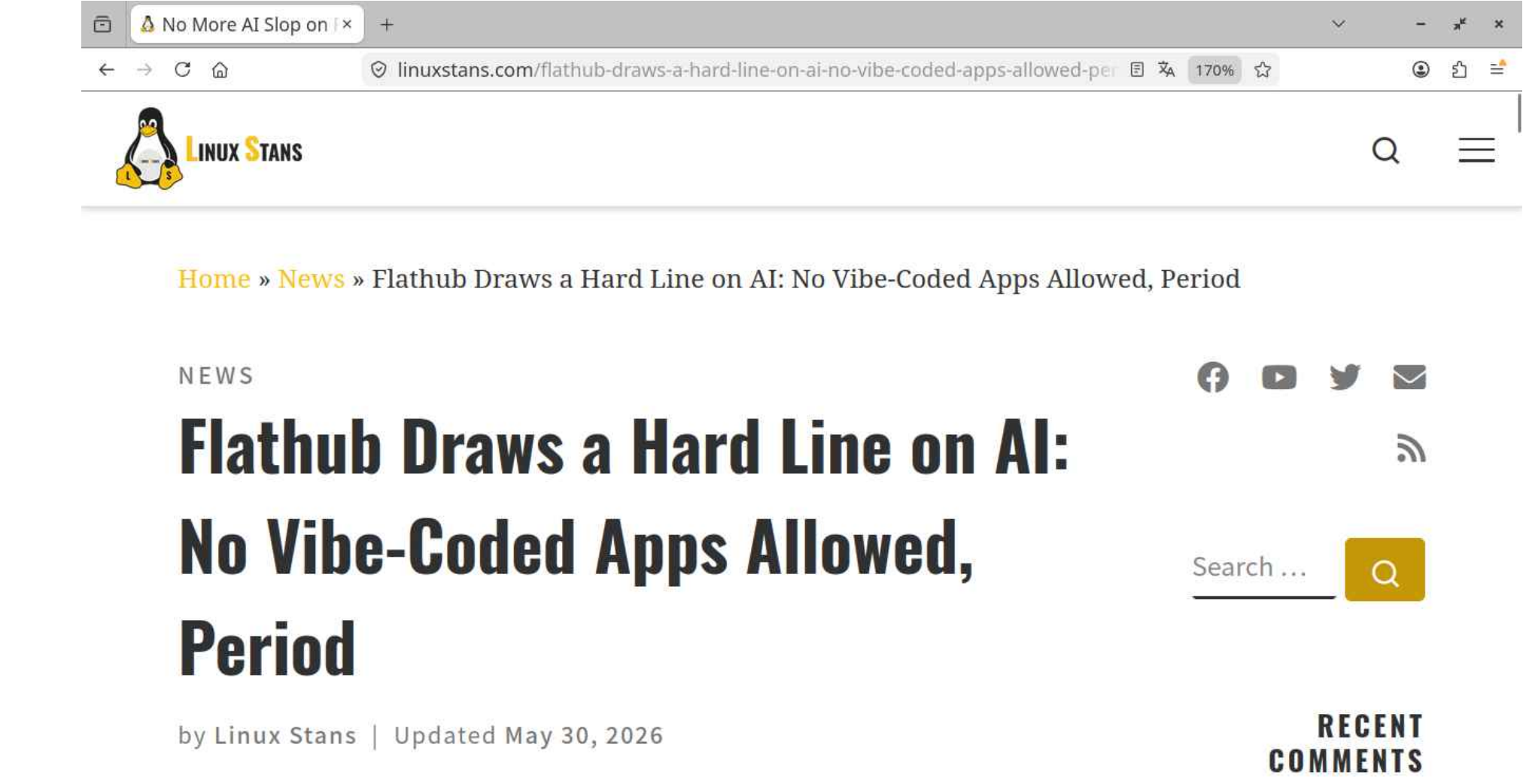
FFmpeg to Google: Fund Us or Stop Sending Bugs

A lively discussion about open source, security, and who pays the bills has erupted on Twitter.

Nov 11th, 2025 8:30am by Steven J. Vaughan-Nichols



https://thenewstack.io/ffmpeg-to-google-fund-us-or-stop-sending-bugs/



https://linuxstans.com/flathub-draws-a-hard-line-on-ai-no-vibe-coded-apps-allowed-period/

Linux 35th Anniversary

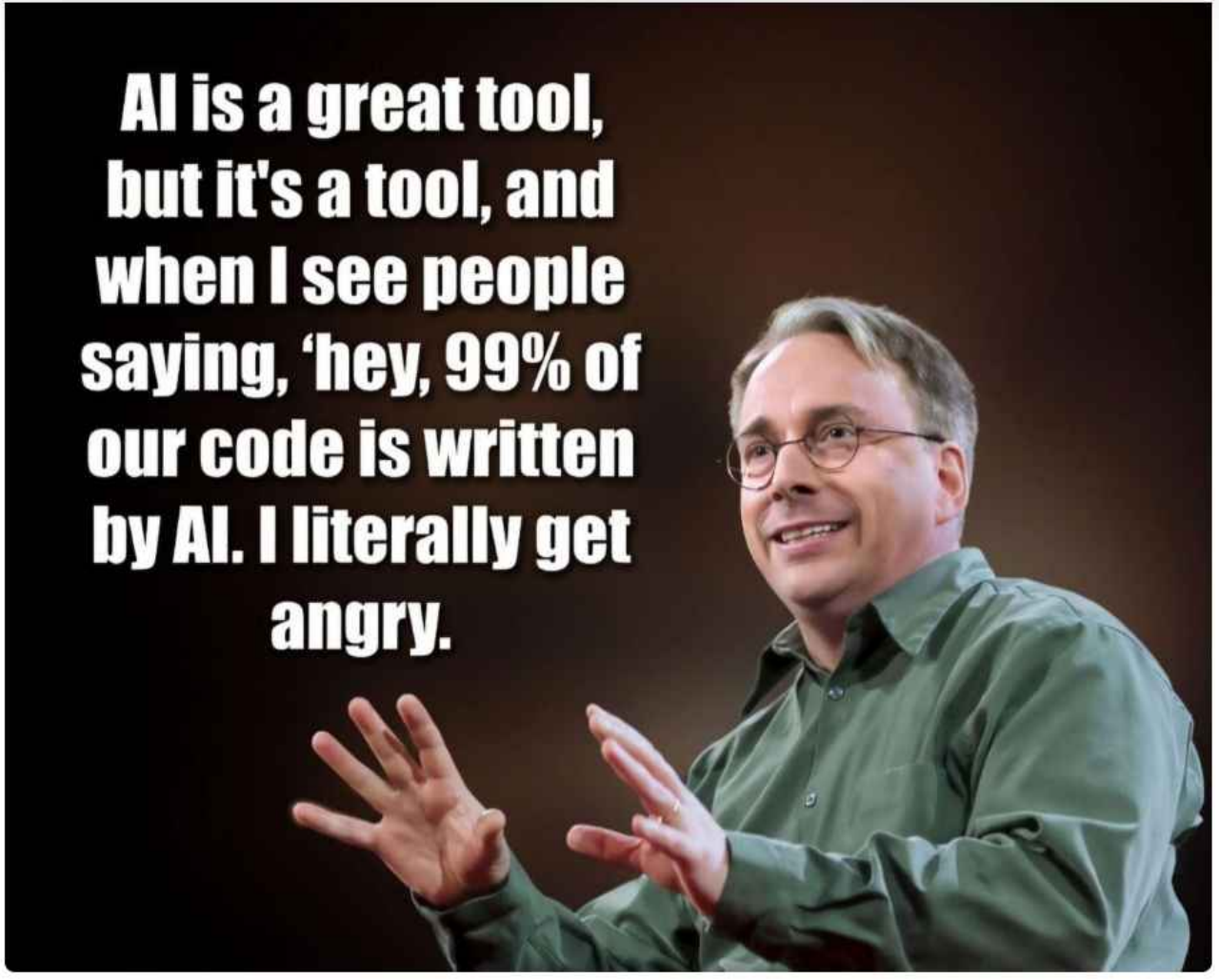


Everything you need to celebrate #Linux35

"I'm doing a (free) operating system (just a hobby, won't be big and professional like gnu)…"

— Linus Torvalds, Usenet, August 25, 1991

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Save these Linux35 graphics and share them wherever you post about #Linux35.
https://www.linux.com/linux35th/



Torvalds shared his views a few months back. Lot has changed since then

https://itsfoss.com/news/torvalds-used-ai-fix-kernel-bug/



linux-media.vger.kernel.org archive mirror

search help / color / mirror / Atom feed

From: Linus Torvalds <torvalds@linuxfoundation.org>
To: Roman Gushchin <roman.gushchin@linux.dev>
Cc: Laurent Pinchart <laurent.pinchart@ideasonboard.com>,
 Mauro Carvalho Chehab <mchehab+huawei@kernel.org>,
Derek Barbosa <debarbos@redhat.com>,
 Matthieu Baerts <matttbe@kernel.org>,
Konstantin Ryabitsev <konstantin@linuxfoundation.org>,
 Jason Gunthorpe <jgg@ziepe.ca>,
Steven Rostedt <rostedt@goodmis.org>,
users@kernel.org,
 Linux Media Mailing List <linux-media@vger.kernel.org>,
Stephen Finucane <stephenfin@redhat.com>
Subject: Re: Linking Patchwork with Sashiko?
Date: Tue, 14 Jul 2026 20:06:14 -0700 [thread overview]
Message-ID: <CAHk-=wi4zC+Ze8e+p3tMv8TtG_80KzsZ1syL9anBtmEh5Z40vg@mail.gmail.com> (raw)
In-Reply-To: <4928C919-7999-4E76-ADCB-F8643FED105B@linux.dev>

On Tue, 14 Jul 2026 at 19:01, Roman Gushchin <roman.gushchin@linux.dev> wrote:
>
> I think it makes the point of sashiko - helping maintainers - unachievable. If the point to not use
> LLMs in general, let's discuss this, not how to make each use case more complex.
>
> It seems like [1] expresses a very anti-LLM position in general

https://lore.kernel.org/linux-media/CAHk-%3Dwi4zC%2BZe8e%2Bp3tMv8TtG_80KzsZ1syL9anBtmEh5Z40vg%40mail.gmail.com/

And no, that's not the position of the Linux kernel.

I realize that some people really dislike AI, but this is an area
where I'm willing to absolutely put my foot down as the top-level
maintainer.

Linux is not one of those anti-AI projects, and if somebody has issues
with that, they can do the open-source thing and fork it.

Or just walk away.

AI is a tool, just like other tools we use. And it's clearly a useful one.

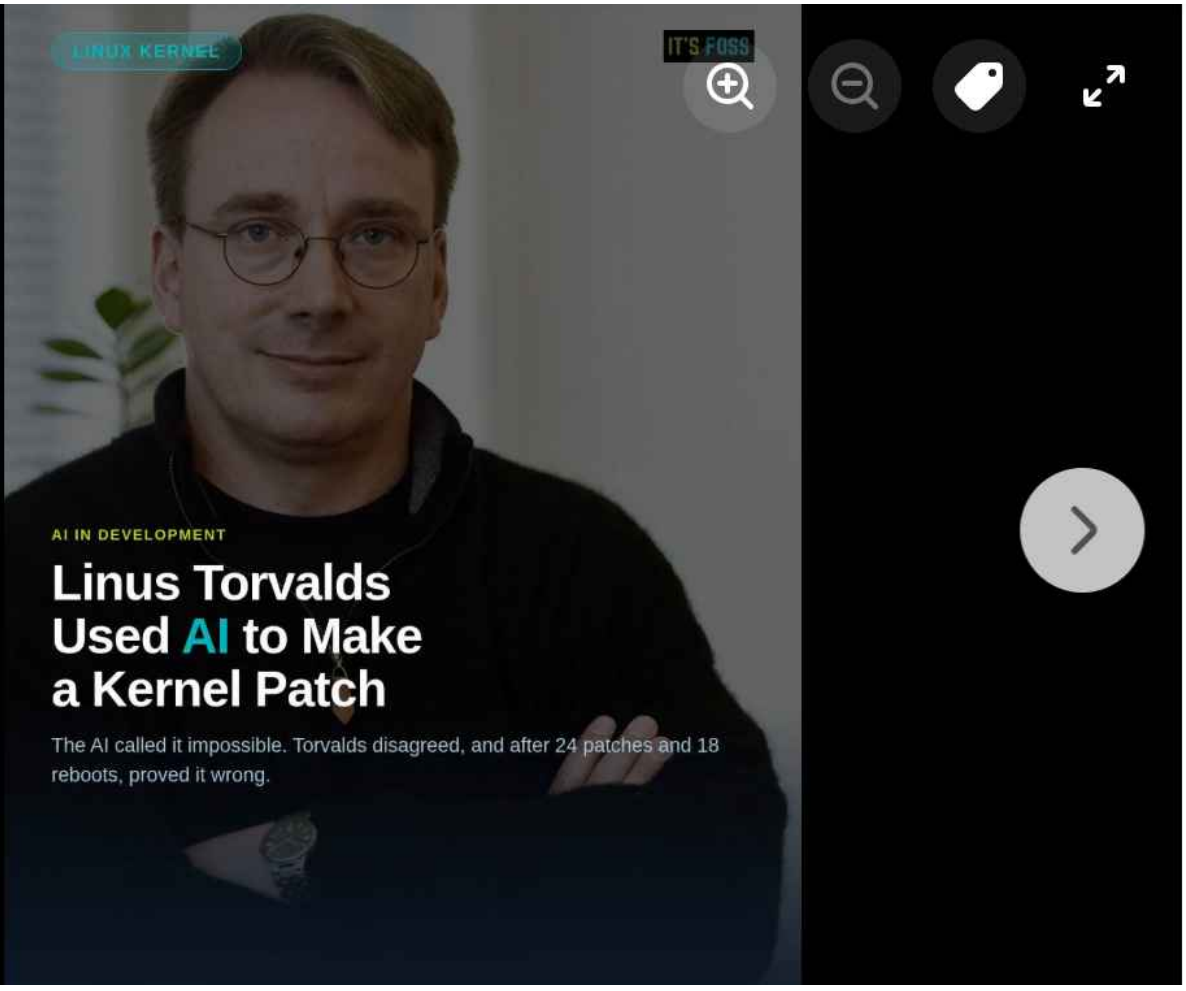
It may not have been that "clearly" even just a year ago, but it's no
longer in question today.

There are other questions around AI (like what the economy of it will
actually look like in the end), but "is it useful" is no longer one of
those questions. Anybody who doubts that clearly hasn't actually used
it.

Yes, it can also be a somewhat painful tool, both for maintainer
workloads and just from a "it keeps finding embarrassing bugs"
standpoint.

But the solution is not to put your head in the sand and sing "La La
La, I can't hear you" at the top of your voice like some people seem

https://lore.kernel.org/linux-media/CAHk-%3Dwi4zC%2BZe8e%2Bp3tMv8TtG_80KzsZ1syL9anBtmEh5Z40vg%40mail.gmail.com/



It's FOSS
8月22日上午10:04 ·

Linus Torvalds just used AI to help fix a Linux kernel bug. And this is a rare event.

Rare because Torvalds doesn't code on Linux kernel anymore. He just reviews code and merge pull requests from contributors.

So this is a rare occasion in recent time where he made a kernel patch himself. And he took assistance of AI this time.

The bug was in the Intel Xe graphics driver for Battlemage G21 cards. It caused GDM, the display manager, to restart endlessly. Tracking it down r…… **查看更多**

https://www.facebook.com/itsfoss/photos/1353924216848628/



https://www.facebook.com/itsfoss/posts/pfbid0HqeQGAdXriQrKrsZdfc5TtbDKS3atozEsrBkjBQr3FPP83NEzk5d7S3NYua1rL7Ml

綜整開源領域的 AI 輔助：

不是全好、或全壞

而是必須佐以適當的管理或規則

掌握與維護人類的核心能力

來盡量讓相關事務有理有據

Google Summer of C ×

www.jenkins.io/projects/gsoc/ai-usage-policy/

Jenkins

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Search CTRL K

Google Summer of Code (GSoC): AI Usage Policy for Jenkins Contributions



At Jenkins, we value authentic contributions, deep understanding, and code quality. While we recognize that AI tools (like ChatGPT, GitHub Copilot, Gemini, Claude, etc.) are powerful aids, their misuse in open source can lead to incorrect code, licensing issues, and maintenance burdens. To ensure the quality of our project, we have established the following guidelines for using AI tools:

For Contributors (or Mentees)

On this page

- For Contributors (or Mentees)
- Why This Policy Exists

Connect

- gitter join chat
- Meetings
- Forum

Recent Blog Posts

- Meet Mohammed Faheem - GSoC 2026 Contributor Working on Jenkins Email Notifications using Outlook SMTP with OAuth
- GSoC 2026 Community

https://www.jenkins.io/projects/gsoc/ai-usage-policy/



LLM Usage

NOTE

⚠️ Failure to disclose LLM usage or submission of unreviewed AI-generated content **may result in proposal dismissal** from GSoC without further review ⚠️

Purpose

This document establishes guidelines for the responsible and transparent use of **AI and Large Language Models (LLMs)** (e.g., ChatGPT, Gemini, Claude, Copilot, etc) by **Google Summer of Code (GSoC)** contributors working on OWASP projects.

The intent of this guideline is to support learning and productivity while preserving originality, accountability, and open-source integrity.

---

1. Acceptable Use

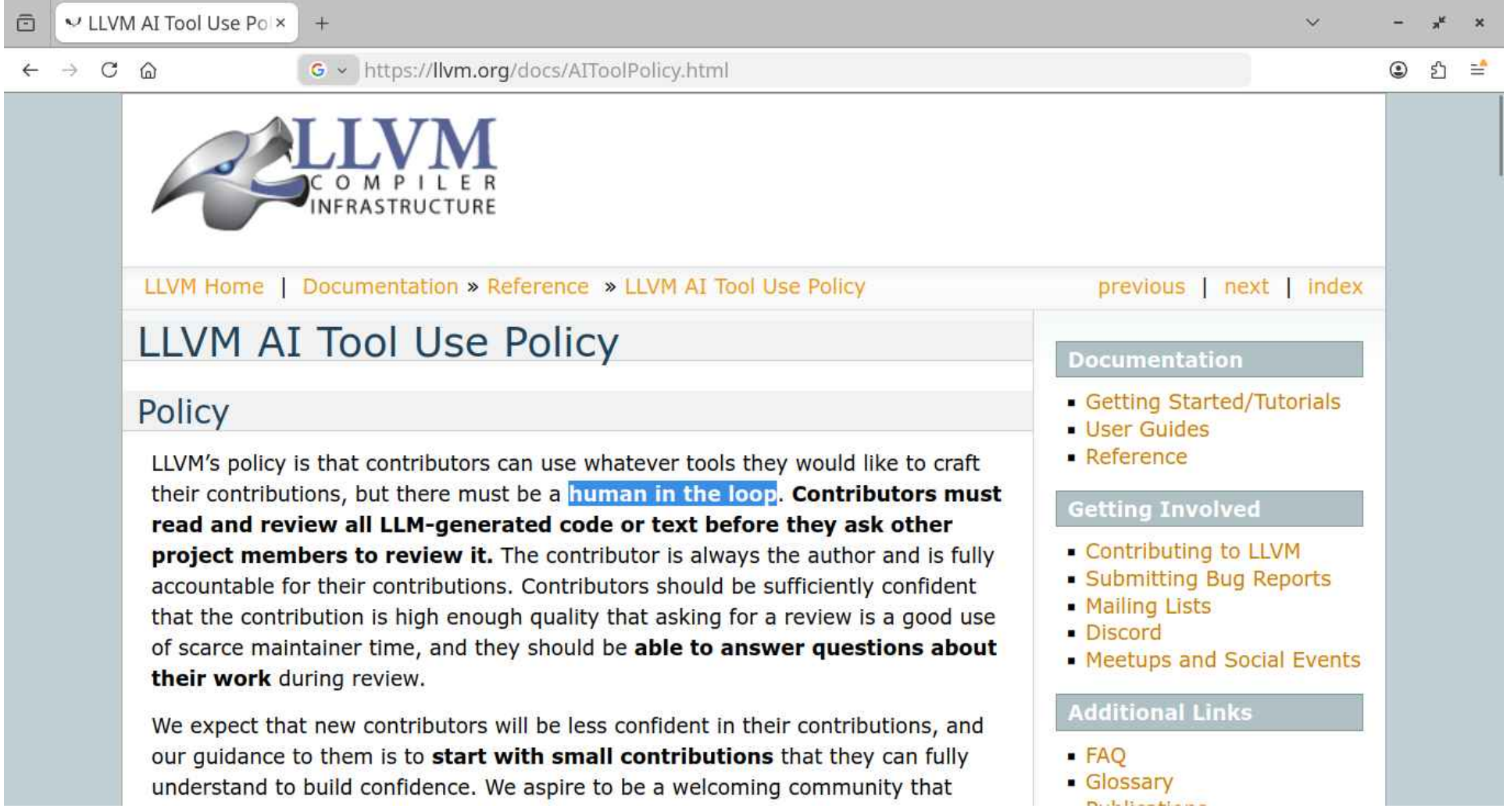
Contributors **may use LLMs** for supportive tasks, including:

- Improving clarity, grammar, or structure of documentation or reports
- Assisting with code comments, summaries, or test-case generation
- Learning and exploration, provided outputs are independently reviewed and understood

https://owasp.org/www-community/initiatives/gsoc/gsoc_ai_usage_guidelines



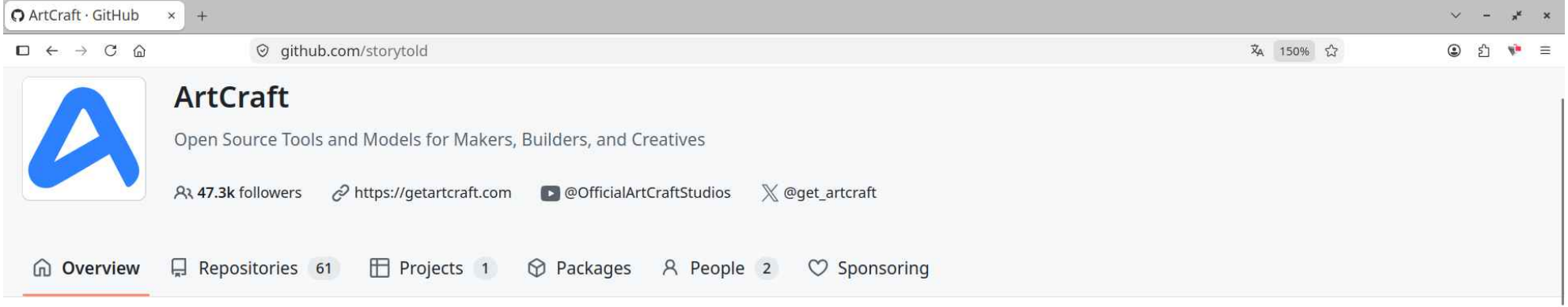
https://genai.owasp.org/



https://llvm.org/docs/AIToolPolicy.html

2.3. 從務實角度得如何因應？

(1) **開源不滅體**、至少近期不會



https://github.com/storytold

Open Core 與 Open Steward 的商業模式和應用分析

開放核心模式（Open Core）

-- 核心開源、增效模組閉源

-- 基礎版本開源、完整版本收費

開放管理者模式（Open Steward）

-- 中立組織不直接營利、協助營利

Decentralized innovation. Built on trust.

The Linux Foundation is a neutral, trusted hub for developers and organizations to code, manage, and scale open technology projects and ecosystems.

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4M+ developers trained ›

855K developers contributing code ›

89M lines of code added weekly ›

21K+ contributing organizations ›

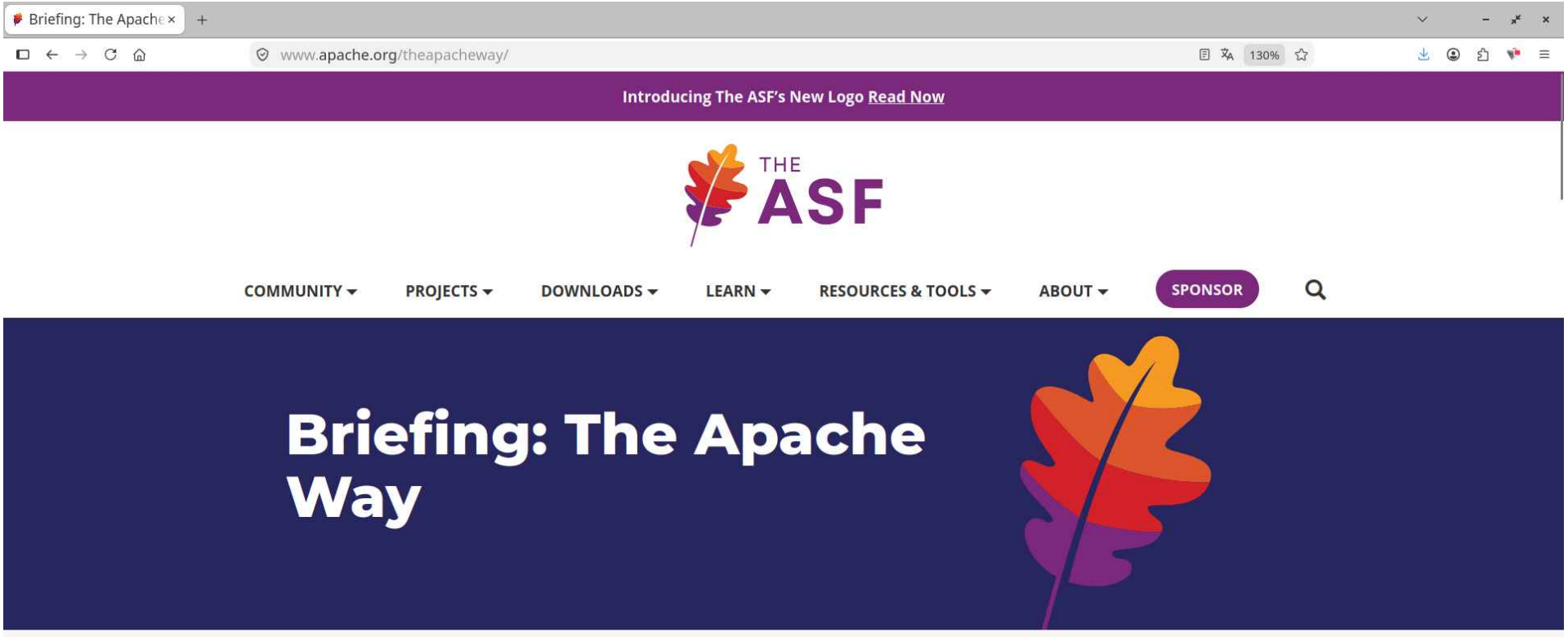
229+ upcoming events ›

100+ standards and specifications ›

https://www.linuxfoundation.org/



https://openssf.org/



https://www.apache.org/theapacheway/



https://news.apache.org/foundation/entry/alc-taipei-igniting-the-apache-open-source-movement-in-taiwan



https://taione.org/gosf/about

開放開源「永續可用、又能更新除錯」的模式

1 、核心開放、由重度商用者付費（ Open Core ）；

2 、由商業社群共同出資出力、交由中立機構託管（ Open Steward ）；

3 、由公益組織或政府機關出錢出力，作為公共建設發展的基礎框架 -- Horizon Europe / Digital Europe Programme 。

Jensen Huang (@Jen

x.com/JensenHuang

200%

Jensen Huang

1 則貼文

回覆

媒體

跟隨

Jensen Huang @JensenHuang · 7月24日

For my first post, I'm sharing a letter @NVIDIA signed on why open models matter.

AI will transform every industry, power every company, and be built by every country.

Open models strengthen safety and cybersecurity, accelerate innovation and diffusion, and enable sovereignty.

顯示更多



Open Weights and
American AI Leadership

July 24, 2026

In the 1980s, early open-source software pioneers challenged the prevailing belief that software would advance only if companies kept tight control over their code. This movement pushed for a transparent ecosystem where developers around the world could study, modify, and improve software. Software developed by the open-source community now supports most of the internet and underlies systems used by the world's largest technology companies, as well as the U.S. military and federal agencies conducting scientific research, cybersecurity, and other critical missions. Open source did more than lower the cost of software; it created a shared foundation of knowledge on which generations of American engineers and entrepreneurs built their institutional sovereignty.

The United States now faces a similar choice with artificial intelligence. Our AI leadership will be judged not by one frontier AI model, but by whether the United States builds a strong, open ecosystem that diffuses into every sector. This is essential for creating opportunities for innovation and prosperity across the country. It requires expanding access to AI, encouraging competition, robust application layers, and giving Americans greater control over the technology they rely on. Open-weight models—AI models that anyone can download, inspect, modify, and run on their own infrastructure—are an important part of that foundation because they make advanced AI more accessible, adaptable, and widely available.

Open weights expand access to the AI economy. Startups, established businesses, universities, and

sovereignty and prosperity.

To be sure, open weights carry real and distinct risks. Once released, the weights are beyond the original developer's control, and modified versions are difficult to trace or reverse. But the right response to this risk is not to prohibit open weights. In a world where cybersecurity attackers use advanced AI, defenders need access to models with comparable capabilities so they can detect, simulate, and respond to emerging threats. Open models broaden defensive capability, increase transparency, and allow vulnerabilities to be discovered and remediated across many teams.

In fact, openness may be one of the most important paths to AI safety and security. Relying solely on closed models is not inherently safe: they can be breached, misused, or fail in ways that outsiders cannot detect. And concentrating advanced AI capabilities behind a small number of closed models compounds that risk. It results in a small number of single points of failure, weakens competition, and leaves critical technology in the hands of a few providers. Open weight models, on the other hand, allow a broad community of researchers and developers to examine their behavior, identify vulnerabilities, develop safeguards, and improve them over time. Just as open-source software demonstrated that transparency can be more secure than obscurity, AI safety may depend on giving more people the ability to test and strengthen the models on which society relies. It allows for rigorous benchmarking and evaluation, red teaming, and protections tied to real and demonstrated harms rather than assuming that closed systems

The age of AI can be one of prosperity. With the right choices, open weight AI can expand opportunity, strengthen competition, extend American technological leadership, mitigate risk, and ensure that the benefits of this extraordinary technology are shared broadly across our economy. That future is worth building, and the United States should lead in building it.

https://x.com/JensenHuang/status/2080643682408321103

X 上的 Jensen Huang

x.com/JensenHuang/status/2080643682408321103/photo/1

200%



Open Weights and
American AI Leadership

July 24, 2026

In the 1980s, early open-source software pioneers challenged the prevailing belief that software would advance only if companies kept tight control over their code. This movement pushed for a transparent ecosystem where developers around the world could study, modify, and improve software. Software developed by the open-source community now supports most of the internet and underlies systems used by the world's largest technology companies, as well as the U.S. military and federal agencies conducting scientific research, cybersecurity, and other critical missions. Open source did more than lower the cost of software; it created a shared foundation of knowledge on which generations of American engineers and entrepreneurs built their institutional sovereignty.

The United States now faces a similar choice with artificial intelligence. Our AI leadership will be judged not by one frontier AI model, but by whether the United States builds a strong, open ecosystem that diffuses into every sector. This is essential for creating opportunities for innovation and prosperity across the country. It requires expanding access to AI, encouraging competition, robust application layers, and giving Americans greater control over the technology they rely on. Open-weight models—AI models that anyone can download, inspect, modify, and run on their own infrastructure—are an important part of that foundation because they make advanced AI more accessible, adaptable, and widely available.

Open weights expand access to the AI economy. Startups, established businesses, universities, and public institutions can build on advanced models without training one from scratch or paying frontier-model prices for every task. Open weights let every organization match the right model to the right job at the right cost, reserving frontier-scale capability for genuine frontier problems and running efficient, specialized models everywhere else. That discipline is what will make AI economically sustainable as its use scales into the billions of everyday tasks. America wins the AI era by diffusing it into the workflows of factories, hospitals, farms, classrooms, and main street businesses.

Open weights also strengthen competition and competition is what keeps the gains of AI broadly shared rather than concentrated in a few hands. By allowing many organizations to build, adapt, and deploy advanced models, open weights create rivalry not only among model developers but across cloud chips,

1.5萬 3.8萬 16萬 5,814萬

Jensen Huang
@JensenHuang

跟隨

顯示翻譯

For my first post, I'm sharing a letter @NVIDIA signed on why open models matter.

AI will transform every industry, power every company, and be built by every country.

Open models strengthen safety and cybersecurity, accelerate innovation and diffusion, and enable sovereignty.

The world needs both frontier closed models and frontier open models.

images.nvidia.com/pdf/Open-Weigh...

https://x.com/JensenHuang/status/2080643682408321103

（中央社洛杉磯24日綜合外電報導）美國人工智慧（AI）晶片巨擘輝達（NVIDIA）執行長黃仁勳今天首度在社群平台X發文，分享一封由輝達和微軟等25家公司連署的公開信，談及開放模型和限制，凸顯美中AI角力。

美國「財星雜誌」（Fortune）網站報導，黃仁勳警告華府不要重蹈過度限制技術的覆轍，就像1980年代也有一些人擔憂開源軟體發展。

這封信主張，開放權重（open-weight）AI模型應作為美國維持領導地位的基石，顯示黃仁勳及其同業確實擔憂這個方向無法實現。

黃仁勳寫道：「開放模型能強化安全性與網路安全，加速創新與普及，實現主權。世界需要前沿閉源模型與前沿開放模型。」



https://www.cna.com.tw/news/aopl/202607250109.aspx



README | Code of conduct | Contributing | Apache-2.0 license | Security

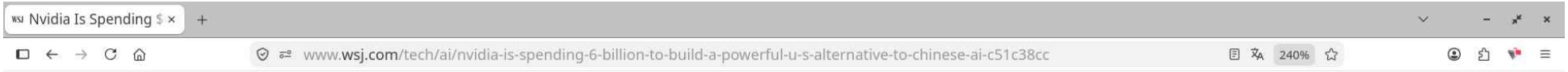
NVIDIA Personal AI Router (PAIR)



NVIDIA Personal AI Router (PAIR) is a local inference router for a group of compatible computers on the same network. It discovers participating nodes, manages supported inference engines, and presents Ollama-compatible and OpenAI-compatible proxy endpoints to applications and agents. Independent requests can be routed to eligible nodes according to engine availability, model availability, and current workload.

PAIR is useful for concurrent local workloads such as multi-agent applications. Prompts and responses are intended to remain on the local network when every

https://github.com/NVIDIA/Personal-AI-Router



THE WALL STREET JOURNAL.

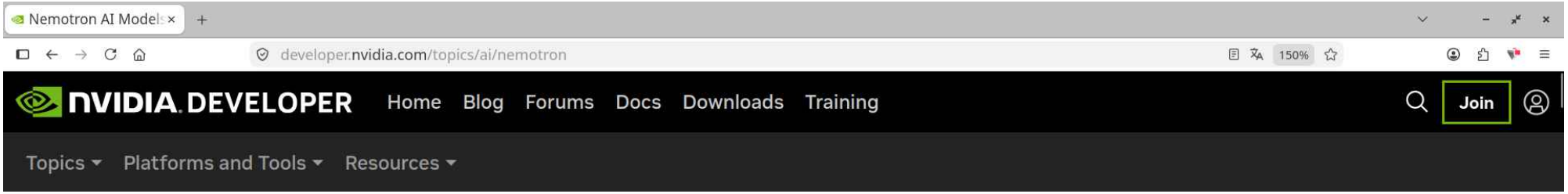
Nvidia Is Spending $6 Billion to Build a Powerful U.S. Alternative to Chinese AI

A sweeping agreement with startup Poolside aims to build an open AI ecosystem in the U.S. to compete with Chinese heavyweights and American AI giants

By Robbie Whelan
Aug. 22, 2026 8:00 pm ET

Aa 298 Listen (2 min)

https://www.wsj.com/tech/ai/nvidia-is-spending-6-billion-to-build-a-powerful-u-s-alternative-to-chinese-ai-c51c38cc



Topics / Artificial Intelligence / NVIDIA Nemotron

On this page

- NVIDIA Nemotron Models
- NVIDIA Nemotron Datasets
- Developer Tools
- Introductory Resources
- Starter Kits
- Inference Service Providers
- More Resources
- Ethical Considerations
- Get Started

NVIDIA Nemotron

NVIDIA Nemotron™ is a family of open models with open weights, training data, and recipes, delivering leading efficiency and accuracy for building specialized AI agents.

Explore Models | **Forum** | **Tutorials**

https://developer.nvidia.com/topics/ai/nemotron

EU Cyber Resilience Act, CRA

Manufacturers shall, upon identifying a vulnerability in a component, including in an open source component, which is integrated in the product with digital elements, report the vulnerability to the person or entity maintaining the component.

IoT and More

Products with digital elements, PDE

法律責任的轉變：在 **CRA** 框架下，將 FOSS 整合進商業產品的「**製造商**」，其法律責任**與過往有何本質上的不同**？


Made with OpenAI GPT-5.6 Sol by Lucien C.H. Lin & Ramus Lin, CC0-1.0.

《優生保健法》

《兒童及少年福利與權益保障法》



https://sutian.moe.edu.tw/zh-hant/su/26974/

IoT / Digital Elements 商品的資安維護（擔保期間的修復、通報、告知）

1. 修復： Expected lifetime ；原則至少 5 年
2. 通報： ENISA ／ CSIRT - Single Reporting Platform

   24hr ／ 72hr ／ 14 days （漏洞）； 1 month （嚴重事件）
3. 告知： Affected users

https://digital-strategy.ec.europa.eu/en/library/commission-publishes-new-guidance-support-timely-cyber-resilience-act-implementation

IoT / Digital Elements 商品的資安維護（擔保期間的修復、通報、告知）

1. 通報： ENISA ／ CSIRT - Single Reporting Platform

   2026.9.11 率先生效

2. CE 標示及其他主要義務： 2027.12.11 全面適用

https://digital-strategy.ec.europa.eu/en/library/commission-publishes-new-guidance-support-timely-cyber-resilience-act-implementation

Single Reporting Platform (SRP)

← Back to main topic

SUBTOPICS

Single Reporting Platform (SRP)

CONTENT

→ Frequently Asked Questions

→ CRA SRP guidance - AR Interface functions

→ CRA SRP guidance - AR Notification submission and update

→ CRA SRP guidance - AR User registration

The Cyber Resilience Act (CRA) introduces the Single Reporting Platform (SRP) for cybersecurity incident reporting in the EU Digital Single Market.

enisa | CRA SINGLE REPORTING PLATFORM

The Single Reporting Platform (SRP) shall become a technical tool for the reporting of actively exploited vulnerabilities and incidents impacting products with digital elements in the EU Digital Single Market.

Under the Cyber Resilience Act (CRA), manufacturers of products with digital elements are mandated to report actively exploited vulnerabilities and severe incidents having an impact on the security of the products.

As of 11 September 2026 onwards, the SRP will be used by CSIRTs and manufacturers for mandatory reporting and could be used by any natural/legal persons for voluntary reporting.

Throughout 2025 and 2026, ENISA is taking the necessary steps to support the successful implementation of the platform.

https://www.enisa.europa.eu/topics/product-security/single-reporting-platform-srp

CRA 對照表

free and open-source software/open-source software steward

Recital 17	對 FOSS 的特別考量
Recital 18	定義 FOSS ，並說明哪些商用面向應受 CRA 管理、哪些純貢獻面向除外。
Recital 19	說明 Open-source Software Steward 制度
Recital 20	說明 GitHub 、 GitLab 等託管平台得除外
Article 3(14)	正式定義 Open-source software steward
Article 24	規定 Open-source software steward 的義務
Article 64(10)	規定 Open-source software steward 不適用行政罰鍰

Article 13 （ Manufacturer obligations ）：商業化提供 FOSS 回歸 CRA 範圍

Article 14 （ Vulnerability handling ）： Steward 須遵循部分漏洞通報義務

軟體生態系（Ecosystem）

-- 以開源為核心協作基礎

-- 善用 AI 工具快速搭建特色功能

-- prompt/agent engineering 的 know-how

為新式護城河

(2) 從容許 / 禁用清單到提升本職學能

企業使用 AI 進行程式開發

(1) 是否允許？

(2) 如何降低風險？

Google有超過1/4的 ×
https://www.ithome.com.tw/news/165758
170%

Google有超過1/4的新程式碼是由AI生成的

Alphabet及Google執行長Sundar Pichai對外揭露該公司的AI基礎設施以及如何利用生成式AI強化旗下服務，其中指出Google的新程式碼中，已經有超過1/4是由AI生成的，再由工程師負責審查與接納

文/ 陳曉莉 | 2024-10-30 發表

讚 355 分享



FAIR USE @ https://www.ithome.com.tw/news/165758

新聞

Gemini 3.5刪除近3萬行程式碼，修改設定，導致用戶系統斷線半小時

有開發人員在Reddit自述，Gemini 3.5在代理式IDE環境中，刪除將近3萬行程式碼，新增錯誤設定並擅自修改專案檔案，導致營運網站斷線超過半小時

文/ 林妍溱 | 2026-05-25 發表



iT+ 看影片追技
AEB大補帖 資安防護大作戰！
資安弱點日新月
曝險管理平台協
站瞭解公司資訊

https://www.ithome.com.tw/news/176087

Github Copilot → Suggestion

(1)telemetry data(程式寫作偵測資料的收集)

(2) 專利演算法

(3)telemetry 功能可關閉但無實質意義

Github Copilot 資通訊產業管理要則 - 早期常見一般規範

(1)Github Copilot 可以用、但必須登錄註記它的使用狀態；

(2) 重點專案要用必須走核可制，所謂重點專案就是涉及公司專利技術，或被指定是主力開發產品，而被定位重點專案，這些專案要使用到 Github Copilot 時，走的是核可制，必須明確得到核可，才可以去使用；

(3) 一般專案走備查制，解放工程人員的生產力，但使用狀況必須於公司指定的電子簿冊進行登錄，相關登錄資訊公司也不需要時時查察，但需要確認時管理和研發主管可以查閱並確認；

(4) 開啟 Github Copilot 提供的 Duplicate Detection 偵測功能，這個功能是 Github Copilot 在美國發生訴訟後， Github 火速加上的新功能，意思就是當 Github Copilot 吐 Suggestions 給軟體開發工程師，工程師也使用了這些被建議的程式碼，之後再加上一個自動化功能，就是幫忙查驗最後的程式碼是否和目前 Github 上託管的主流他人專案具有表達的近似性，若是高度相似，這個偵測功能會建議部份程式碼再另行改寫，以避免程式碼表達上過於近似有抄襲疑慮。

(5) 要求開發人員就最後提交給公司的程式碼進行人力的閱讀和確認。

Human in the Loop

Human on the Loop



https://www.youtube.com/watch?v=vGxpFgHo1uo

AI 程式輔助開發工具
提供應用相關擔保嗎

Github Copilot/ Claude Code/ OpenAI Codex

1. 免費版 - 沒有！

2. 付費版 - 有限度的有

(1) 第三方請求抗辯機制

(2) 限定範圍補償機制

但若做這些事或涉入這些範圍、不賠！

(1) 第三方請求抗辯機制

- 出事沒講、和解不找、隱瞞事實

(2) 限定範圍補償機制

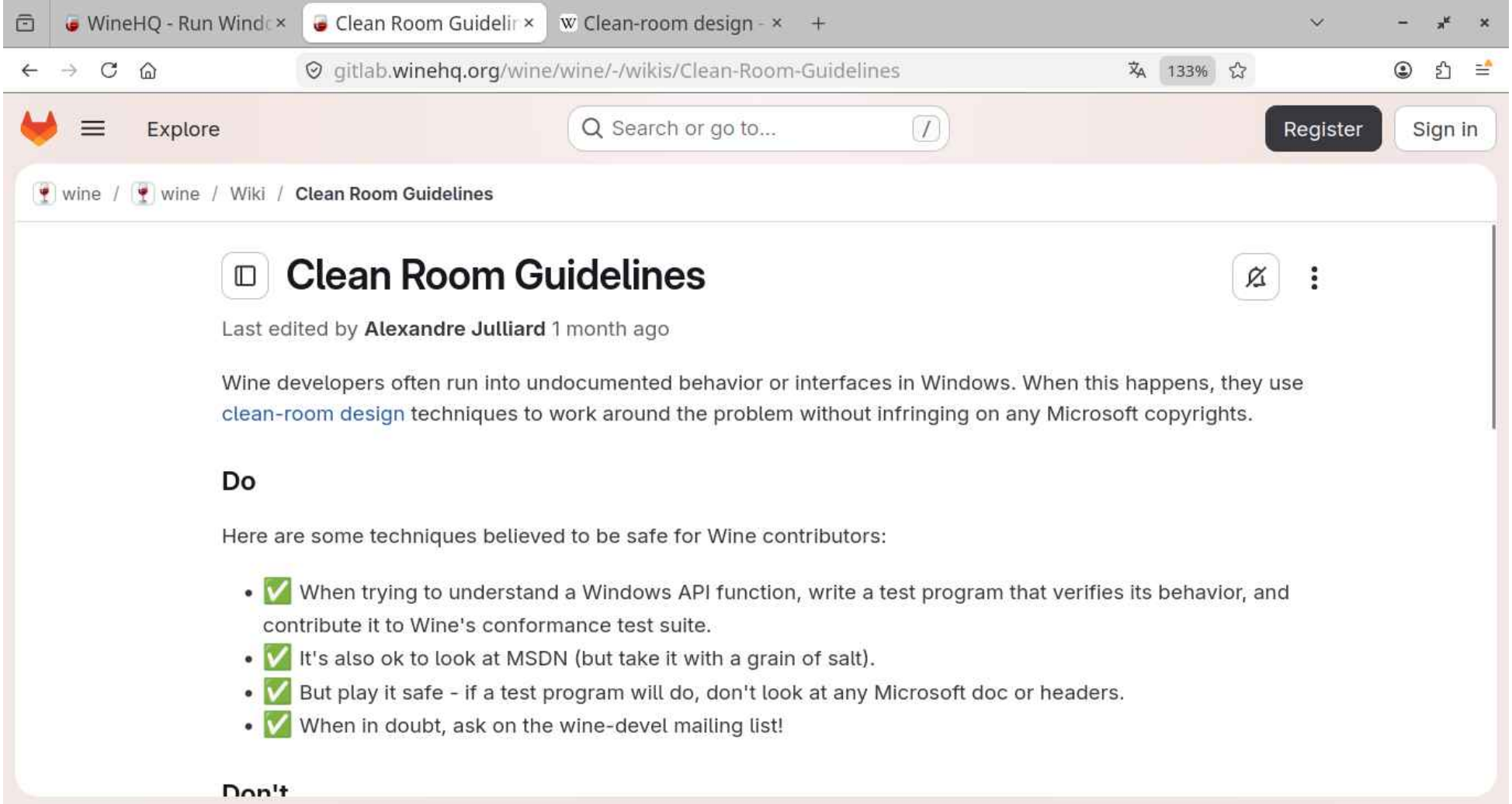
- 可歸責於客戶（明知違法、自行抄襲、不開偵測）

→ 程序紀錄與版本控制系統非常重要

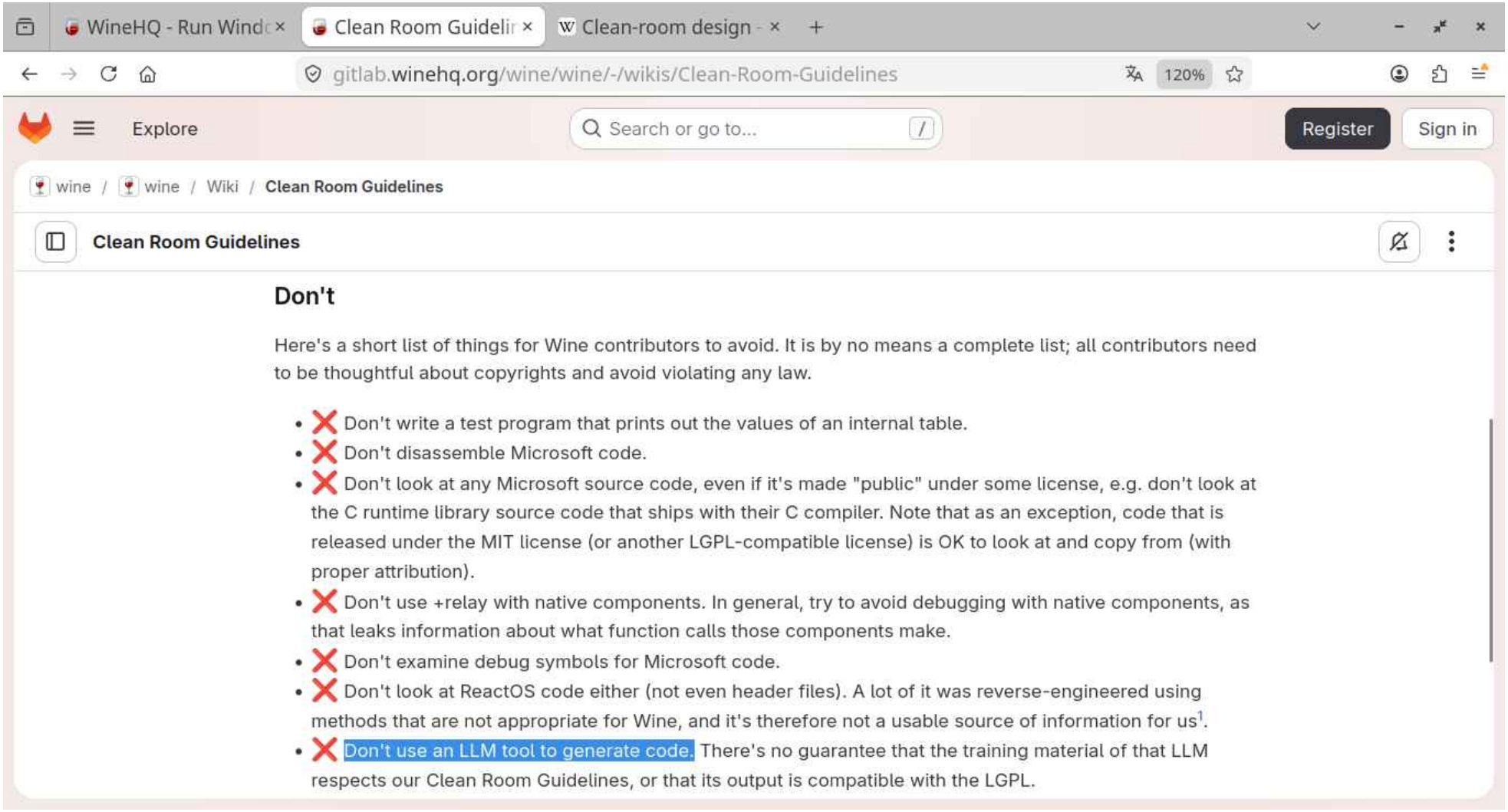
(3) WINE Clean Room 的借鏡



https://www.winehq.org/



https://gitlab.winehq.org/wine/wine/-/wikis/Clean-Room-Guidelines



https://gitlab.winehq.org/wine/wine/-/wikis/Clean-Room-Guidelines

WineHQ Clean Room Guidelines DO

1. 當需要瞭解某個 Windows API 函式的具體行為時，應優先撰寫測試程式進行驗證，並將相關成果納入 Wine 的相容性測試套件。
2. 參考 MSDN 文件原則上並無不可，但其內容未必完全準確，仍應審慎評估。
3. 為降低法律與技術污染風險，若能透過測試程式確認行為，即不建議再查閱 Microsoft 的官方文件或標頭檔。
4. 如對適當作法有所疑慮，請於 wine-devel 郵件論壇中與社群討論。

https://gitlab.winehq.org/wine/wine/-/wikis/Clean-Room-Guidelines

WineHQ Clean Room Guidelines Don't

1. 不要撰寫僅用來列印（ print ）內部資料表（ internal table ）內容的測試程式。
2. 不要對 Microsoft 的程式碼進行反組譯（ disassembly ）。
3. 不要查閱任何 Microsoft 原始碼，即使該原始碼係依某種授權條款公開釋出。例如，不要查閱隨 Microsoft C 編譯器附帶之 C Runtime Library 原始碼。但有一項例外：若相關程式碼係以 MIT License 或其他與 LGPL 相容之授權條款釋出，則可在適當標示來源（ attribution ）的前提下查閱及引用其內容。
4. 不要對原生（ native ）元件使用 +relay 除錯模式。一般而言，應盡可能避免透過原生元件進行除錯，因為這可能洩漏該等元件實際呼叫了哪些函式，進而取得不應取得的實作資訊。
5. 不要檢視 Microsoft 程式碼的除錯符號（ debug symbols ）。
6. 也不要查閱 ReactOS 的程式碼（甚至連標頭檔都不要）。 ReactOS 有相當部分程式碼係透過不適用於 Wine 的逆向工程方法所產生，因此對 Wine 專案而言，該等程式碼不應作為可接受的資訊來源。
7. 不要使用大型語言模型（ LLM ）工具來產生程式碼。因為無法保證該 LLM 的訓練資料符合 Wine 的 Clean Room Guidelines ，也無法保證其輸出內容與 LGPL 授權條款相容。

https://gitlab.winehq.org/wine/wine/-/wikis/Clean-Room-Guidelines

WineHQ Clean Room Guidelines Don't

不要使用大型語言模型（LLM）工具來產生程式碼。因為無法保證該 LLM 的訓練資料符合 Wine 的 Clean Room Guidelines ，也無法保證其輸出內容與 LGPL 授權條款相容。

WineHQ Clean Room Guidelines Don't

所以必須看過 MS 源碼或其高度關聯資訊（除錯碼）才能寫得出來的東西，都可以被視為觀察為著作表達形式本身，而無法成功主張 CLEAN ROOM ！

請 AI 協助條列分析程式碼要點

→ 概念、事實資訊之學習不受著作保護或限制

→ 而非直接下令轉譯他人程式、承襲其表達

知識可以學習、表達不宜直接完全承襲！

1. 分析的結果是否單純為**知識點** / **互動邏輯**
2. 原程式的**表達是否仍在**？
3. 會否構成為原軟體的**改作**？
4. 指示先就原軟體架構解構後進行**架構學習**！

版本控制— Version Control

版本控制（英語： Version control ）是維護工程藍圖的標準做法，能追蹤工程藍圖從誕生一直到定案的過程。此外，版本控制也是一種軟體工程技巧，藉此能在軟體開發的過程中，確保由不同人所編輯的同一程式檔案都得到同步。

https://zh.wikipedia.org/zh-tw/版本控制

AI 程式寫作的合規應用

AI Compliance Engineering

1.公司政策：容許列表 / 拒絕列表

2.請 AI 找 bug 、補洞、請 AI 資安弱掃→工具性使用

3.不要用 AI 整隻轉譯第三方程式碼

4.必要時請 AI 分析其互動邏輯、再依此邏輯重新寫程式

5.毋須所有程式重新寫、 MIT 或寬鬆的 APIs 直接用

6.做好版本控制紀錄的保存以供必要驗證

(4) 歐盟的 AI 標示義務－以 Claude Code 的調整為說明範例

AI 輔助開發應否揭露？

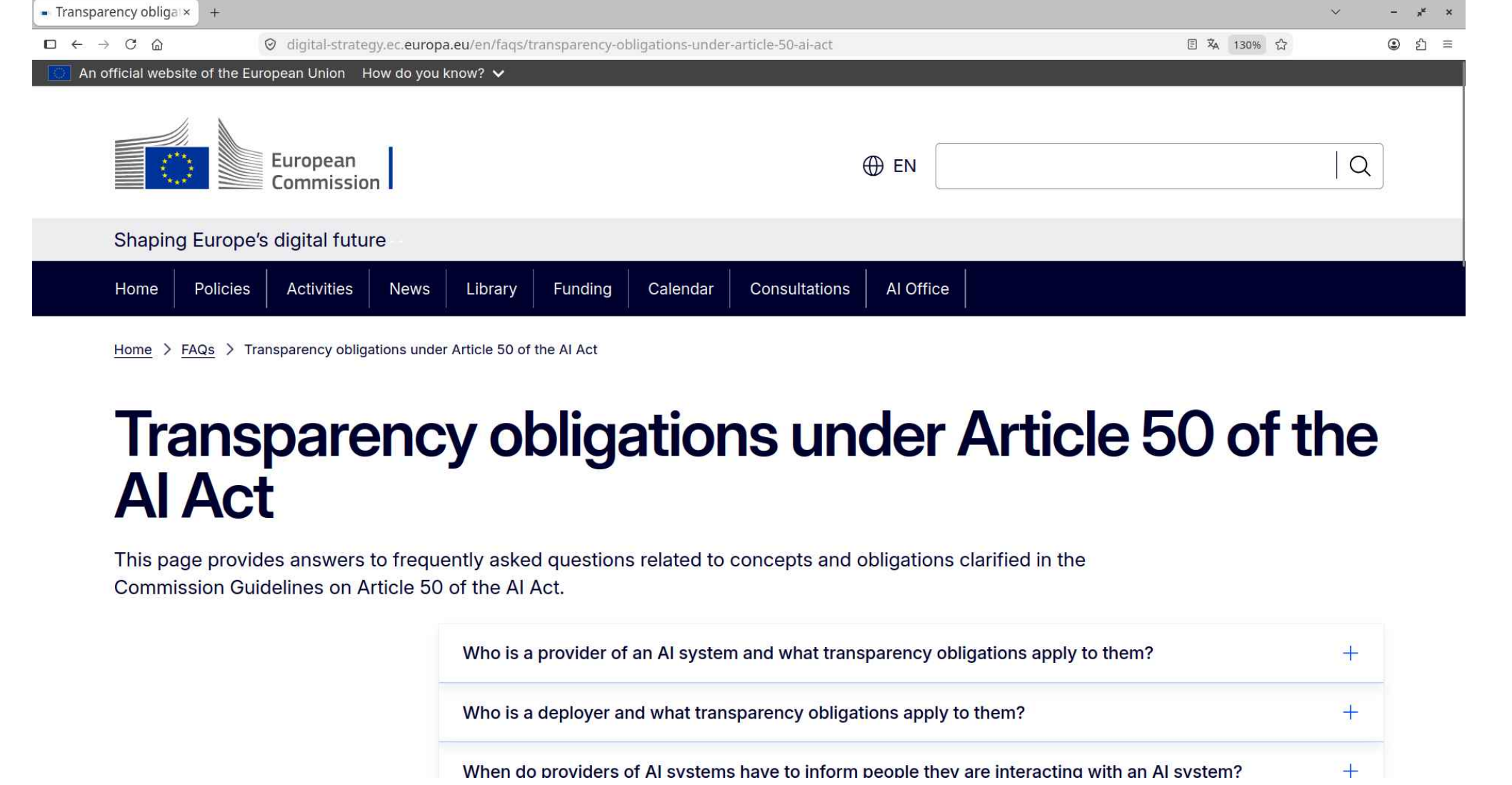
向上級必須揭露

開發歷程應登錄

發布時概負其責視為人類作品

EU AI ACT 要求避免誤會

Article 50 2026.8.2 正式上路



https://digital-strategy.ec.europa.eu/en/faqs/transparency-obligations-under-article-50-ai-act



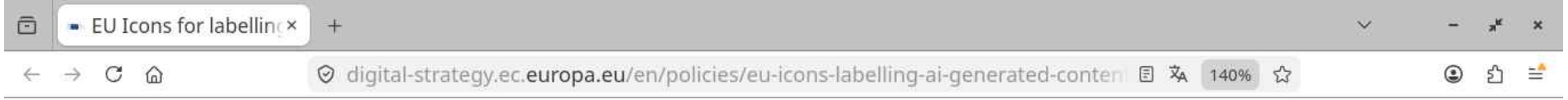
What content needs to be labelled as AI-generated or manipulated?

The icons are intended to help natural persons exposed to such content to recognise, in a clear and distinguishable manner, that the content has been artificially generated or manipulated. This obligation supports public trust, reduces the risk of misinformation and the informed viewing and reading by natural persons of all sorts of content.

Not all AI-generated or manipulated content needs to be labelled. The disclosure requirement under the AI Act covers only AI-generated or manipulated content. The icons support compliance with Article 50(4) of the AI Act, which requires deployers of AI systems to disclose:

- **Deep fakes** - AI-generated or manipulated image, audio, or video content that resembles existing persons, objects, places, entities or events and would falsely appear to a person to be authentic or truthful.
- **AI-generated or manipulated text** published to inform the public on matters of public interest, which did

https://digital-strategy.ec.europa.eu/en/policies/eu-icons-labelling-ai-generated-content



Limitations and exceptions

Evidently artistic, creative, satirical, fictional and analogous works

Where deep fake content forms part of an evidently artistic, creative, satirical, fictional or analogous work or programme, the transparency obligations are limited to disclosure in an appropriate manner that does not hamper the display or enjoyment of the work.

Authorised by law

The disclosure obligation does not apply where the use of deepfakes or published text on matters of public interest is authorised by law to detect, prevent, investigate or prosecute criminal offences.

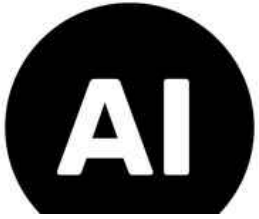
Human review or editorial control for text

https://digital-strategy.ec.europa.eu/en/policies/eu-icons-labelling-ai-generated-content

The EU icons

The Icons have 4 variations: black, white, black with 50% transparency and white with 50% transparency. You can download zip files with all the icons in all variations in SVG and PNG formats.

The icons have undergone user-testing, and the results have informed their design. Notably, performance improved across all measures when the basic icon was accompanied by a text label (e.g. modified).

Icon	When to use	Use case examples
Basic icon 	When AI was involved in the creation of deep fake content (image, audio, video) or published text or when a custom text label or interactive second layer is implemented	Deep fake video with the text label “voices generated with” followed by the basic icon

https://digital-strategy.ec.europa.eu/en/policies/eu-icons-labelling-ai-generated-content

EU Icons for labellin ×
digital-strategy.ec.europa.eu/en/policies/eu-icons-labelling-ai-generated-conten
140%

Fully AI-Generated 	When the entire deep fake content (image, audio, video) or text is fully generated by AI with no human-created elements or human editorial control (apart from prompting)	Fully AI-generated deep fake videos with politicians or fictional events Fully AI-composed music or art* AI-generated news summaries
Partially AI-Modified 	When pre-existing, human-made content was partially modified with AI turning it into a deep fake or text on matters of public interest	A face in an authentic photograph is swapped with a face of a politician using AI Authentic photographs of an empty apartment is furnished

https://digital-strategy.ec.europa.eu/en/policies/eu-icons-labelling-ai-generated-content

人工智慧基本法

第 16 條

數位發展部應參考國際標準或規範，推動與國際介接之人工智慧風險分類框架，並應協助各目的事業主管機關訂定以風險為基礎之管理規範。

各目的事業主管機關應視人工智慧應用風險管理之需要，循前項風險分類框架，訂定以風險為基礎之管理規範，並應協助相關產業自行訂定產業指引及行為規範。有第五條第一項所列之情事者，應依法令限制或禁止之。

人工智慧基本法

第 5 條

政府應避免人工智慧之應用，有侵害人民生命、身體、自由或財產，破壞社會秩序、國家安全或生態環境，或偏差、歧視、廣告不實、資訊誤導或造假等違反相關法規之情事。

政府應以兒少最佳利益為原則，人工智慧產品或系統經中央目的事業主管機關會商數位發展部認定為高風險應用者，應明確標示注意事項或警語。

數位發展部及其他相關機關應提供或建議評估驗證之工具或方法，以利各目的事業主管機關辦理前項事項。

前項驗證工具及方式之形成，應徵詢相關利益團體、產業、學者、社會團體及法律專家之意見。

現階段 AI 產出標示的建議原則 -v.20260804

1. **應予標示**：不揭露者可能**直接違反現行法**（例如觸及 刑法、選罷法、醫師法 相關法律保留事項之 deepfaking）
2. **建議標示**：涉及**風險分擔**、**公平原則**的其他事項

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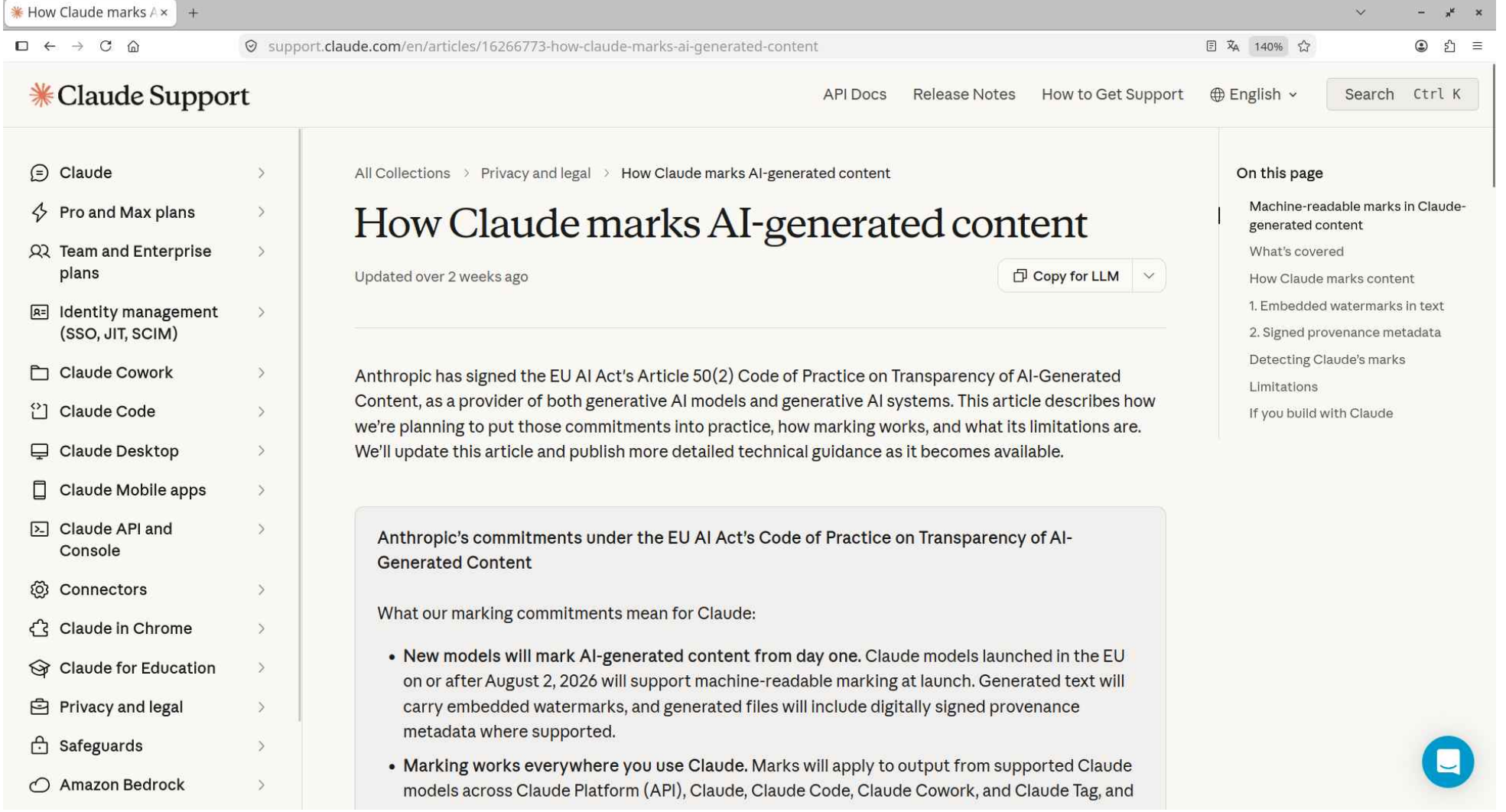
3. **不標亦可**：然標示有助於**釐清或聲明權利**

目漸形成的實務情境（流程化 / 自動化）：

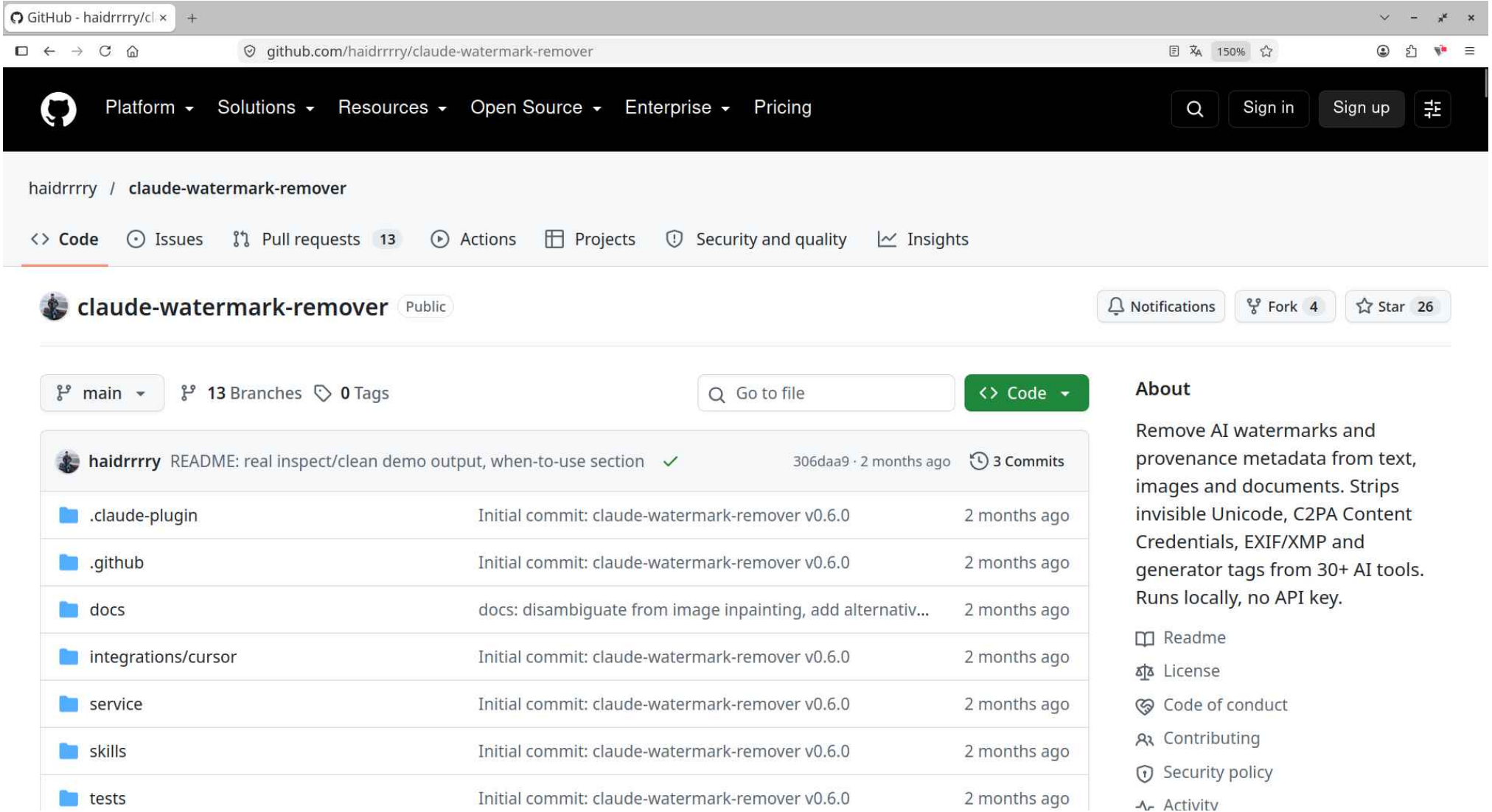
1 、 AI 防偽標示（沒去刪會自動在那裡）： Adobe C2PA 、 OpenAI DALL・E 、 Stability AI invisible watermark ， Google Studio SynthID 。

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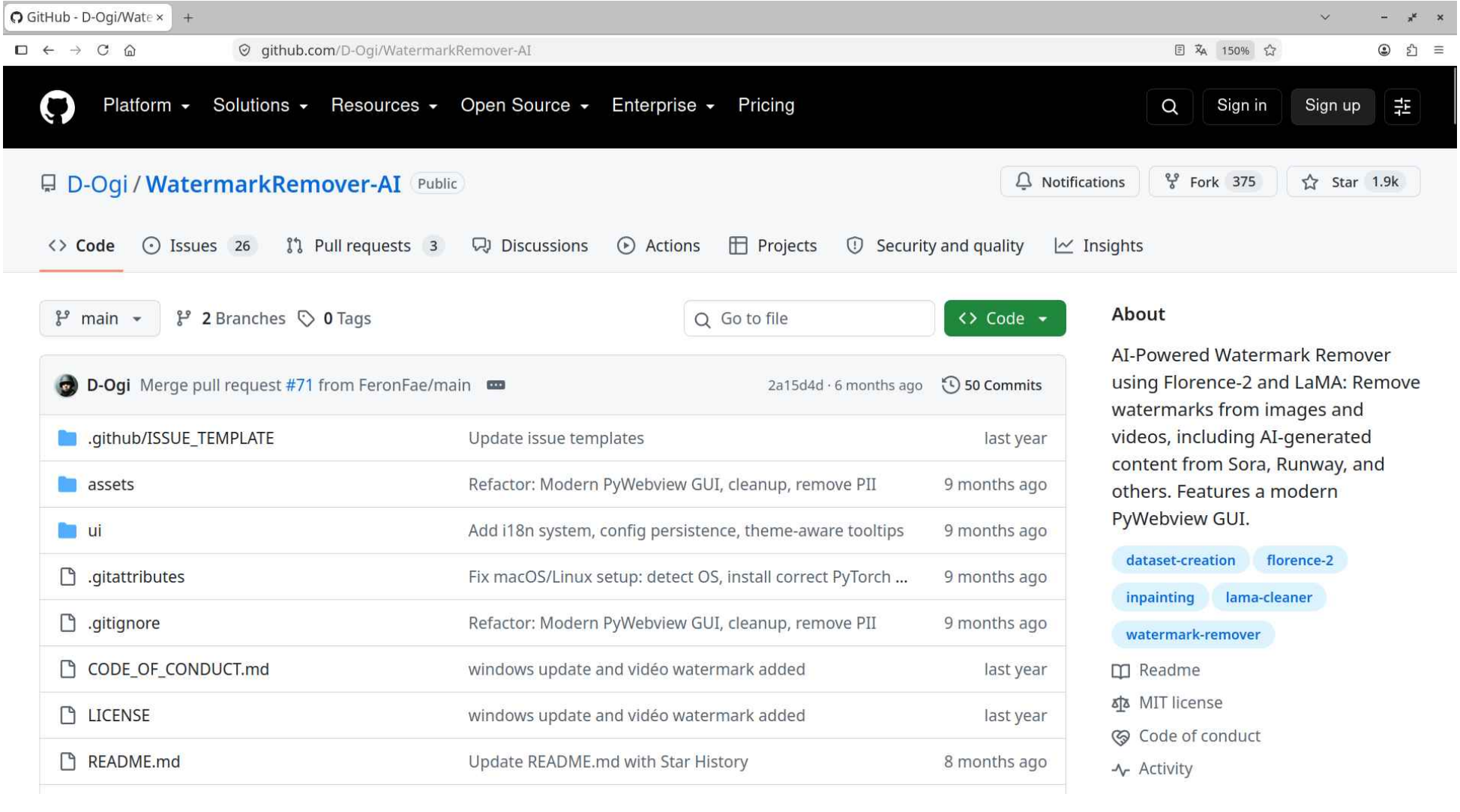
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https://support.claude.com/en/articles/16266773-how-claude-marks-ai-generated-content



https://github.com/haidrrrry/claude-watermark-remover

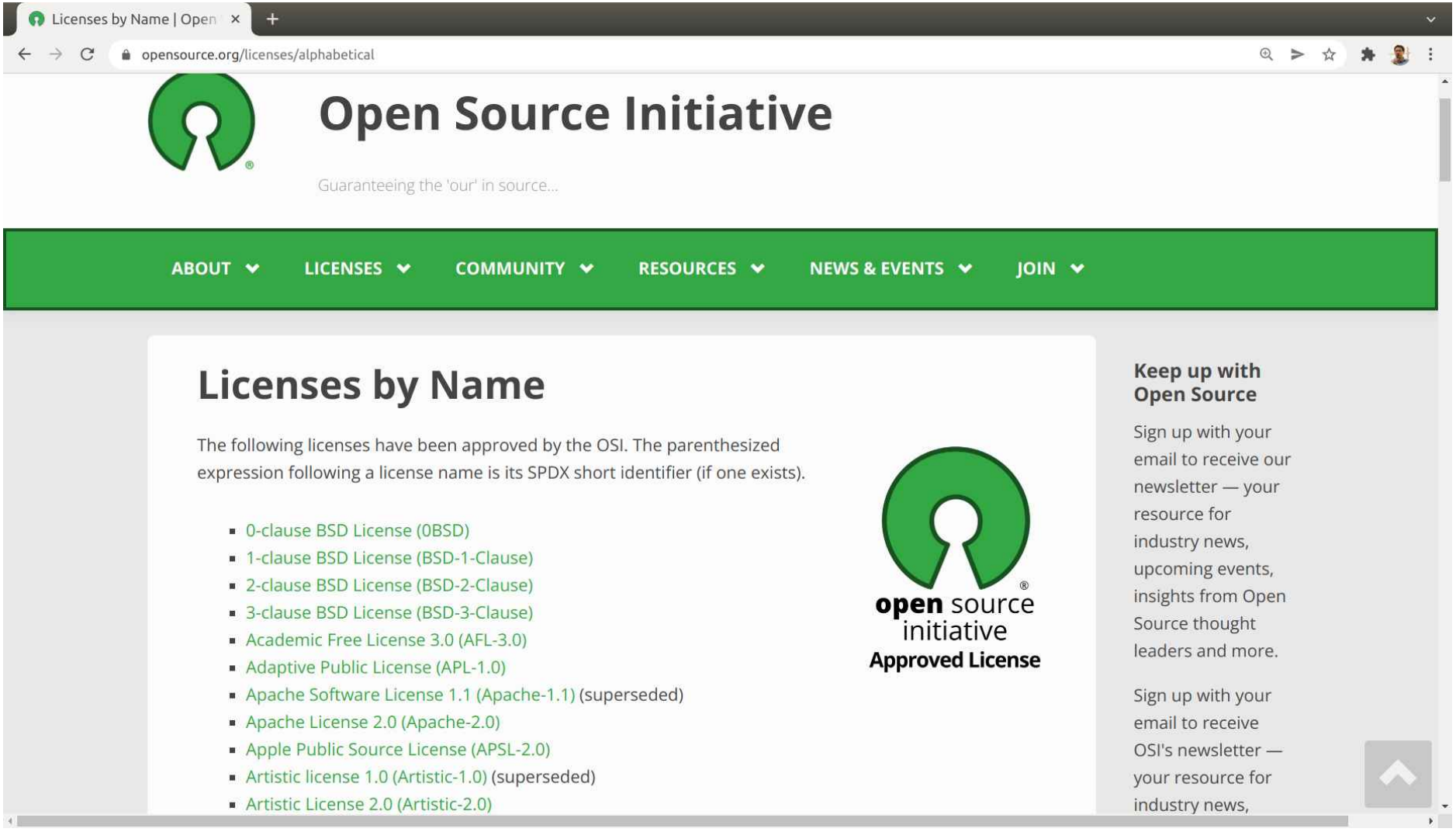


https://github.com/D-Ogi/WatermarkRemover-AI

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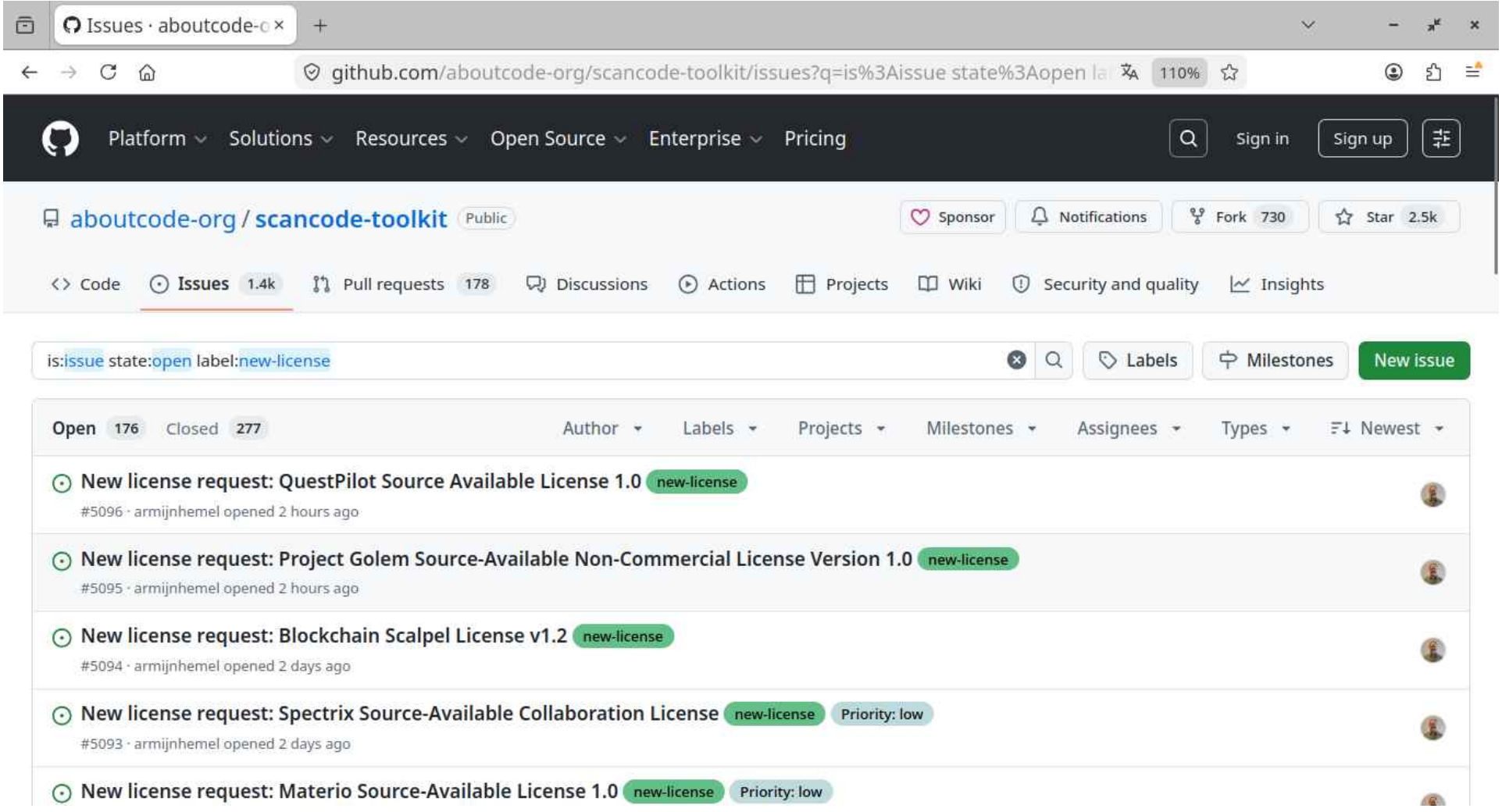
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